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Transfer ramp systems for loading, unloading, or transferring cargo between one or more commercial vehicles and associated systems, devices, and methods

Abstract

Transfer ramp systems, including transfer ramp systems for loading, unloading, or transferring cargo between one or more commercial vehicles, are disclosed herein. In one embodiment, a transfer ramp system comprises a transfer ramp and at least one extendable arm attached to the transfer ramp. The at least one extendable arm is usable to extend or retract the transfer ramp. The transfer ramp system is mountable beneath a top platform of a vehicle or trailer. When the transfer ramp system is mounted beneath the top platform, the transfer ramp is moveable between (a) a stowed position in which the transfer ramp is positioned beneath the top platform and (b) a deployed position in which the transfer ramp is usable to bridge a gap between the top platform and another structure for moving cargo between (i) the other structure and (ii) the vehicle or trailer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims the benefit of the following applications: (a) U.S. Provisional Patent Application No. 63/499,977, filed May 3, 2023; (b) U.S. Provisional Patent Application No. 63/583,151, filed Sep. 15, 2023; (c) U.S. Provisional Patent Application No. 63/585,202, filed Sep. 25, 2023; and (d) U.S. Provisional Patent Application No. 63/566,210 filed Mar. 15, 2024. (2) All of the foregoing applications are incorporated herein by reference in their entireties. Further, components and features of embodiments disclosed in the applications incorporated by reference may be combined with various components and features disclosed and claimed in the present application.

TECHNICAL FIELD
(1) The present disclosure relates generally to transfer ramp systems. For example, several embodiments of the present technology relate to transfer ramp systems for loading and unloading one or more commercial vehicles, including for transferring cargo between two or more commercial vehicles.
BACKGROUND
(2) The trucking industry serves an essential function of an economy's supply chain by transporting goods and other materials over land, typically from a port or warehouse to either retail distribution centers or consumers' residential addresses. Commonly, specialty equipment, such as forklifts and docks, are used to load and unload commercial shipping vehicles. In addition, transferring cargo between commercial vehicles remains challenging even with known specialty equipment and human supervision and/or intervention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present disclosure. The drawings should not be taken to limit the disclosure to the specific embodiments depicted, but are for explanation and understanding only.
(2) FIG. **1** is a partially schematic perspective view of a transfer ramp system configured in accordance with various embodiments of the present technology.

(3) FIG. **2** is another partially schematic perspective view of the transfer ramp system of FIG. **1**.
(4) FIGS. **3**A-**3**D are partially schematic cross-sectional side views of the transfer ramp system of FIG. **1**.
(5) FIG. **3**E is a partially schematic side view of the transfer ramp system of FIG. **1**.
(6) FIGS. **4**A and **4**B are partially schematic perspective views of sliding rails of extendable arms of the transfer ramp system of FIG. **1** configured in accordance with various embodiments of the present technology.
(7) FIG. **5** is a partially schematic perspective view of an actuator of the transfer ramp system of FIG. **1** configured in accordance with various embodiments of the present technology.
(8) FIGS. **6**A-**6**C are partially schematic perspective views of a cable latch system of the transfer ramp system of FIG. **1** configured in accordance with various embodiments of the present technology.
(9) FIGS. **7**A and **7**B are partially schematic perspective views of a strut of the transfer ramp system of FIG. **1** configured in accordance with various embodiments of the present technology.
(10) FIGS. **8**A-**8**L are partially schematic side views of the transfer ramp system of FIG. **1** illustrating a method of operating the transfer ramp system in accordance with various embodiments of the present technology.
(11) FIGS. **9**A-**9**C are partially schematic side views of another transfer ramp system configured in accordance with various embodiments of the present technology.
(12) FIGS. **10**A-**10**C are partially schematic cross-sectional side views of still another transfer ramp system configured in accordance with various embodiments of the present technology.
(13) FIGS. **11**A and **11**B are partially schematic side views of yet another transfer ramp system configured in accordance with various embodiments of the present technology.
(14) FIGS. **12**A-**12**D are partially schematic side views of still another transfer ramp system configured in accordance with various embodiments of the present technology.
(15) FIGS. **13**A-**13**E are partially schematic side views of still another transfer ramp system configured in accordance with various embodiments of the present technology.
(16) FIG. **14** is a partially schematic cross-sectional side view of yet another transfer ramp system configured in accordance with various embodiments of the present technology.
(17) FIG. **15**A is a partially schematic cross-sectional side view of another transfer ramp system configured in accordance with various embodiments of the present technology.
(18) FIGS. **15**B and **15**C are partially schematic side views of a variation of the transfer ramp system of FIG. **15**A configured in accordance with various embodiments of the present technology.
(19) FIGS. **16**A-**16**E are partially schematic perspective views of another transfer ramp system configured in accordance with various embodiments of the present technology.
(20) FIGS. **17**A-**17**C are partially schematic perspective views of still another transfer ramp system configured in accordance with various embodiments of the present technology.
(21) FIGS. **18**A and **18**B are partially schematic perspective views of another transfer ramp system configured in accordance with various embodiments of the present technology.
(22) FIGS. **18**C and **18**D are partially schematic, front views of a portion of the transfer ramp system of FIGS. **18**A and **18**B.
(23) FIGS. **19**A and **19**B are partially schematic perspective views of another portion of the transfer ramp system of FIGS. **18**A-**18**D.
(24) FIG. **20** is a perspective view of the transfer ramp system of FIGS. **18**A-**18**D in a deployed state.
(25) FIG. **21** is a partially schematic side view illustrating motion of the transfer ramp system of FIGS. **18**A-**18**D.
(26) FIGS. **22**A-**22**N are partially schematic side views of the transfer ramp system of FIGS. **18**A-**18**D illustrating a method of operating the transfer ramp system in accordance with various embodiments of the present technology.

(27) FIGS. **23**A-**23**N are other partially schematic side views of the transfer ramp system of FIGS. **18**A-**18**D illustrating the method of operating the transfer ramp system shown in FIGS. **22**A-**22**N.
(28) FIGS. **24**A-**24**N are partially schematic, cross-sectional side views of the transfer ramp system of FIGS. **18**A-**18**D illustrating the method of operating the transfer ramp system shown in FIGS. **22**A-**22**N.
(29) FIGS. **25**A-**25**D are perspective views of a transfer ramp stowage system configured in accordance with various embodiments of the present technology.
(30) FIGS. **26**A-**26**E are partially schematic views of another transfer ramp stowage system configured in accordance with various embodiments of the present technology.
(31) FIG. **27** is a partially schematic perspective view of a curved track configured in accordance with various embodiments of the present technology.
(32) FIG. **28** is a partially schematic perspective view of a user controls arrangement configured in accordance with various embodiments of the present technology.
(33) FIGS. **29**A and **29**B are partially schematic side perspective views of a transfer ramp system configured in accordance with various embodiments of the present technology.
(34) FIG. **30** is a partially schematic diagram of a controls system configured in accordance with various embodiments of the present technology.
(35) FIGS. **31**A and **31**B illustrate a flow diagram showing a process for controlling a transfer ramp system in accordance with various embodiments of the present technology.
(36) FIG. **32** is a partially schematic diagram of a diagnostics board or module configured in accordance with various embodiments of the present technology.
(37) FIG. **33** is a partially schematic perspective view of a travel limiter on an actuator of a transfer ramp system configured in accordance with various embodiments of the present technology.
(38) FIGS. **34**A-**34**C are partially schematic perspective views of a stowage stop of a transfer ramp system configured in accordance with various embodiments of the present technology.

DETAILED DESCRIPTION

A. Overview

(39) Shipping companies (e.g., less-than-truckload (LTL) expeditors) frequently transfer cargo (e.g., pallets or other objects) between commercial vehicles. For example, when a truck transports cargo to an airport for shipping the cargo elsewhere, that truck must commonly wait in line at the airport for one to three hours to drop off the cargo before being freed up to perform additional work. Thus, when a company simultaneously employs (i) a first truck to pick up a first pallet from a first location and (ii) a second truck to pick up a second pallet from a second location, the second pallet in the second truck is commonly transferred to the first truck before the first truck heads to the airport to drop off both the first and second pallets. This truck-to-truck transfer frees up the second truck to perform additional work, while only the first truck must wait at the airport for one to three hours. In other words, in such a scenario, the truck-to-truck transfer facilitates realizing one to three hours of additional utilization of the second truck.
(40) Today, such truck-to-truck transfers are typically performed by positioning trucks back-to-back and jumping a gap that exists between the trucks. More specifically, rubber bumpers on a liftgate of a first truck can be backed against rubber bumpers on a liftgate of a second truck. In this position, there commonly exists a gap between the dock plate of the liftgate of the first truck and the dock plate of the liftgate of the second truck. This gap is typically about 26 cm (e.g., 25.4 cm, or 10 inches) wide. Thus, to transfer a pallet from the second truck into the first truck, the drivers employ a pallet jack to jump the gap and move the pallet into the first truck. A similar transfer operation is commonly performed at a loading dock, such as (i) when the loading dock lacks a dock plate that spans a gap between a top surface of the loading dock and a liftgate of a truck and/or (ii) when a mechanism used to deploy the dock plate of the loading dock is inoperable.
(41) There are several drawbacks to performing transfers of cargo in this manner. For example, using a pallet jack to jump the gap is a physically intensive procedure for drivers of the commercial

vehicles. In addition, such transfers are typically limited to cargo weighing approximately 136-182 kg (approximately 300-400 lbs.) or less, which excludes a large percentage of cargo typically transferred by such commercial vehicles. Furthermore, such a transfer procedure poses a risk (i) of damaging one or both of the commercial vehicles, the pallet jack, the cargo being transferred across the gap, and/or other cargo in one or both of the commercial vehicles; (ii) of injuring the drivers; (iii) of the wheels of the pallet jack or other objects falling into or getting stuck within the gap; and/or (iv) of the cargo falling (e.g., falling over, falling off the pallet jack, falling off the liftgates, and/or falling into the gap) while jumping the gap.

(42) To address these concerns, several embodiments of the present technology are directed to transfer ramp systems (also referred to herein as "cargo transferring systems") for loading, unloading, and/or transferring cargo between (a) a commercial vehicle and/or (b) another commercial vehicle, loading dock, trailer, or other structure. In one embodiment, for example, a transfer ramp system includes a ramp coupled to a pair of extendable arms. The arms are actuatable by one or more actuators to move the ramp from a stowed position beneath a top platform (e.g., of or above a liftgate) at a rear of a commercial vehicle to an extended position above a top platform. The transfer ramp system can further include a release latch that, when actuated, allows the ramp to pivot into alignment with the top platform (e.g., freely, via use of one or more struts, via use of one or more springs, via use of an electric motor or other actuator, and/or by manually rotating the ramp).

(43) After the ramp is pivoted into alignment with the top platform, the one or more actuators are used to retract the extendable arms such that the ramp is brought into contact with (a) the top platform and (b) a top platform of another commercial vehicle, a top surface of a loading dock, or a top surface of another structure. In this manner, the transfer ramp system can be used to bridge the gap that exists between (i) the end of the top platform of a commercial vehicle on which the transfer ramp system is installed and (ii) the end of a top platform/surface of another commercial vehicle, a loading dock, or another structure. As such, cargo (e.g., of any weight) can be more easily and safely transferred into, out of, and/or between one or more commercial vehicles, without needing to jump the gap and/or with less risk (i) of damaging one or both of the commercial vehicles, a pallet jack, the cargo being transferred, and/or other cargo in one or both of the commercial vehicles; (ii) of injuring the driver(s); and/or (iii) of the cargo falling over and/or off the liftgate(s) of the commercial vehicle(s).

(44) Specific details of several embodiments of transfer ramp systems (and associated systems, devices, and methods) are described below. For example, several embodiments of the present technology described below are directed to transfer ramp systems for transferring cargo between (a) a commercial vehicle and (b) another commercial vehicle, loading dock, or other structure. Although the present technology is primarily described below in relation to commercial vehicles (e.g., commercial trucks, semi-trucks, trailers, etc.), a person of ordinary skill in the art will readily appreciate that other vehicles (e.g., personal, recreational, civilian, etc.) can be used with the systems, devices, methods, and computer-readable mediums of the present technology described herein.

(45) It will be appreciated that variations of the embodiments illustrated in the drawings exist and are within the scope of the present technology. Furthermore, it should be noted that embodiments of the present technology can have different configurations, components, and/or procedures than those shown or described herein. Moreover, a person of ordinary skill in the art will understand that embodiments of the present technology can have configurations, components, and/or procedures in addition to those shown or described herein, and that these and other embodiments can lack several of the configurations, components, and/or procedures shown or described herein without deviating from the present technology.

B. Selected Embodiments of Transfer Ramp Systems and Associated Systems, Devices, and Methods

(46) FIG. **1** is a partially schematic perspective view of a transfer ramp system **100** ("the system **100**") configured in accordance with various embodiments of the present technology. As shown, the system **100** includes a transfer ramp **102** (e.g., a transfer platform, a transfer bridge, a transfer deck, a bridge deck), a plurality of extendable arms **104** (identified individually in FIG. **1** as first extendable arm **104***a* and second extendable arm **104***b*), mounting brackets **106** (identified individually in FIG. **1** as first mounting bracket **106***a* and second mounting bracket **106***b*), and a ramp latching system **110**. Although not shown in FIG. **1**, the system **100** can additionally include (a) one or more actuators for extending and/or retracting the first and second extendable arms **104***a* and **104***b*, and/or (b) one or more struts and/or springs that facilitate rotating or pivoting the transfer ramp **102**. Such actuators, struts, and/or springs are discussed in greater detail below with reference to FIGS. **2**, **5**, and **7**A-**8**L.
(47) In FIG. **1**, the system **100** is illustrated in a stowed (e.g., undeployed, retracted, fully retracted) position or state beneath a liftgate **120** of a vehicle or trailer (e.g., a commercial vehicle, a commercial trailer, a box truck). More specifically, the liftgate **120** has a top platform **122** (e.g., a top plate, a dock plate) and sides **126**. The sides **126** are identified individually in FIG. **1** as first side **126***a* and second side **126***b*, and each includes steps **125** and a rubber bumper **127** (e.g., used as a stop while backing the liftgate **120** against a loading dock or another vehicle or trailer). In the stowed position, the system **100** occupies a space beneath the top platform **122** and between the first and second sides **126***a* and **126***b* of the liftgate **120**. The first and second mounting brackets **106***a* and **106***b* of the system **100** of FIG. **1** are fixedly attached (i) to a portion of the liftgate **120** beneath the top platform **122** and/or (ii) to the vehicle/trailer such that the system **100** is mounted to or under/beneath frame rails of the vehicle/trailer. The frame rails can include frame rails of the frame of the vehicle/trailer itself and/or frame rails of a frame of a box positioned on or carried by the vehicle/trailer. As a specific example, the first and second mounting brackets **106***a* and **106***b* of the system **100** can be fixedly attached to the portion of the liftgate **120** beneath the top platform **122** and/or to the vehicle/trailer such that the system **100** is mounted to, on, within, and/or beneath a chassis of the vehicle/trailer. As shown in FIGS. **1**, the first and second mounting brackets **106***a* and **106***b* are also attached to the first and second extendable arms **104***a* and **104***b*, respectively. In turn, the first and second extendable arms **104***a* and **104***b* are each attached to a corresponding side of the transfer ramp **102**.
(48) The ramp latching system **110** of the system **100** includes a cable release latch **111** that releasably holds the transfer ramp **102** in the orientation shown in FIG. **1**. Such an orientation of the transfer ramp **102** is also referred to herein as a 'stowed orientation.' The ramp latching system **110** further includes an actuation mechanism **115** and a cable **112** operably connecting the actuation mechanism **115** to the cable release latch **111**. The actuation mechanism **115** can include a handle, button, lever, or other user interface that, when pushed, pulled, twisted, or otherwise manipulated by a user of the system **100**, releases the cable release latch **111** via the cable **112** to permit the transfer ramp **102** to pivot forward from the stowed orientation toward an orientation in which the transfer ramp **102** is parallel to or aligned with the top platform **122** of the liftgate **120** (e.g., an orientation in which the transfer ramp **102** is generally flat or horizontal). As best shown in FIGS. **2**, **3**E, and **8**A-**8**L, the cable release actuation mechanism **115** can be mounted to or stowed at/on a portion of the liftgate **120** in some embodiments, such as to the second side **126***b* of the liftgate **120**. Additionally, or alternatively, the cable release actuation mechanism **115** can be mounted and/or stowed at a location that is proximate controls for the liftgate **120**. Furthermore, although shown with the cable release latch **111** in the illustrated embodiment, the ramp latching system **110** can include a different style of latch (i) that releasably holds the transfer ramp **102** in the stowed orientation and (ii) that may or may not be actuated using the actuation mechanism **115** and/or the cable **112** shown in FIG. **1**. Various different styles of latches and latching systems are discussed in greater detail below.
(49) FIG. **2** is a partially schematic perspective view of the transfer ramp system **100** of FIG. **1**

illustrated in a deployed state (e.g., a fully deployed state). As shown, the transfer ramp **102** of the system **100** has been (i) extended out from beneath the top platform **122** of the liftgate **120** via the extendable arms **104***a* and **104***b*, (ii) pivoted forward from the stowed orientation to a generally horizontal or flat orientation, and (iii) lowered via the extendable arms **104***a* and **104***b* until the bottom surface **102***b* (FIG. **1**) of the transfer ramp **102** abuts against the top platform **122** of the liftgate **120** and a top surface **232** of another structure **230**. In some embodiments, the other structure **230** can be a liftgate of another commercial vehicle, and the top surface **232** can be a top platform (e.g., a dock plate) of the liftgate of the other commercial vehicle. In these and other embodiments, the other structure **230** can be a trailer (e.g., a semi-truck trailer) or a bed of a truck, and the top surface **232** can be a floor or other surface of the trailer or the bed of the truck. Additionally, or alternatively, the other structure **230** can be a loading dock, and the top surface **232** can be a top surface of the loading dock. In these and still other embodiments, the other structure **230** can be another transfer ramp system (e.g., similar to the system **100**), and the top surface **232** can be a top surface of a transfer ramp of the other transfer ramp system.

(50) In the deployed state illustrated in FIG. **2**, the transfer ramp **102** spans or bridges a gap that exists between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**. More specifically, a first portion of the transfer ramp **102** rests on the top surface **232** of the other structure **230**, and a second portion of the transfer ramp **102** rests on the top platform **122** of the liftgate **120**. As such, the transfer ramp **102** facilitates transferring cargo between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230** via a top surface **102***a* of the transfer ramp **102**. In other words, the transfer ramp **102** facilitates loading cargo into and/or unloading cargo from the vehicle/trailer, such as without needing to back the top platform **122** of the liftgate **120** fully against the top surface **232** of the other structure **230**.

(51) Because the transfer ramp **102**, in the deployed state illustrated in FIG. **2**, rests on the top platform **122** of the liftgate **120** and on the top surface **232** of the other structure **230**, weight of cargo positioned on the top surface **102***a* of the transfer ramp **102** can be transferred from the transfer ramp **102** to the liftgate **120** and the other structure **230**. Thus, at least while the system **100** is in the deployed state shown in FIG. **2**, the weight of cargo positioned on the top surface **102***a* of the transfer ramp **102** can be supported by the transfer ramp **102**, the liftgate **120**, the vehicle/trailer employing the liftgate **120**, and the other structure **230**. In other words, in such a state of the system **100**, the weight of the cargo on the top surface **102***a* of the transfer ramp **102** need not be supported by the extendable arms **104***a* and **104***b* and/or the first and second mounting bracket **106***a* and **106***b* of the system **100**. Instead, assuming nothing is placed on the transfer ramp **102** outside of the deployed state shown in FIG. **2**, the extendable arms **104***a* and **104***b* and/or the first and second mounting brackets **106***a* and **106***b* need only support the weight of the transfer ramp **102** while deploying or retracting the transfer ramp **102** between the stowed position (FIG. **1**) and the deployed position (FIG. **2**).

(52) As shown in FIG. **2**, the transfer ramp **102** can include a back lip **208***a* and a front lip **208***b*. The back lip **208***a* can correspond to a backmost edge of the transfer ramp **102** relative to a front of the vehicle/trailer, and/or the front lip **208***b* can correspond to a frontmost edge of the transfer ramp **102** relative to the front of the vehicle/trailer. The back lip **208***a* and/or the front lip **208***b* can be ramped to facilitate a smoother or gradual transition between (a) the top surface **232** of the other structure **230** and/or the top platform **122** of the liftgate **120** and (b) the top surface **102***a* of the transfer ramp **102**. The smoother or gradual transition can facilitate easier loading and/or unloading of cargo onto and/or off of the top surface **102***a* of the transfer ramp **102**.

(53) In these and other embodiments, the transfer ramp **102** can include side rails **203** (identified individually in FIG. **2** as first side rail **203***a* and second side rail **203***b*). In some embodiments, the first and second side rails **203***a* and **203***b* are positioned to help prevent or hinder cargo from rolling or otherwise shifting off the top surface **102***a* of the transfer ramp **102** into the gap between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**. In some

embodiments, the first and second extendable arms **104***a* and **104***b* can be mechanically attached to the first and second side rails **203***a* and **203***b*, respectively, such that the first and second extendable arms **104***a* and **104***b* can raise and/or lower the transfer ramp **102** as the first and second extendable arms **104***a* and **104***b* are extended and/or retracted, respectively. As discussed above, the system **100** can include struts **207***a* and **207***b* (only the strut **207***a* is illustrated in FIG. **2**; the strut **207***b* is shown in FIG. **3**B). The struts **207***a* and **207***b* can be mechanically attached to the first and second side rails **203***a* and **203***b*, respectively, of the transfer ramp **102**, and can be used to control or limit tilt or pivot of the transfer ramp **102** (e.g., at least when the transfer ramp **102** is not held in the stowed orientation by the cable release latch **111** of the ramp latching system **110**). Furthermore, the struts **207***a* and **207***b* and/or the first and second extendable arms **104***a* and **104***b* can prevent the transfer ramp **102** from shifting and/or rotating along a plane generally parallel to the top platform **122** of the liftgate **120**, at least while the transfer ramp **102** is in the deployed position illustrated in FIG. **2**.

(54) FIGS. **3**A-**3**E illustrate a process of operating the system **100** to move the system **100** from the stowed position (as illustrated in FIG. **1**) to the deployed position (as illustrated in FIG. **2**). More specifically, FIGS. **3**A-**3**D are partially schematic cross-sectional side views of the transfer ramp system **100** taken along the line **3**A-**3**A in FIG. **1**, and FIG. **3**E is a partially schematic side view of the transfer ramp system **100** without the cross-section. In FIG. **3**A, the system **100** is illustrated in the stowed position beneath the top platform **122** of the liftgate **120**. In FIG. **3**B, the extendable arms **104***a* (FIG. **1**) and **104***b* of the system **100** are extended such that the transfer ramp **102** is moved in a direction generally parallel to arrow A and through the gap that exists between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**.

(55) In FIG. **3**C, after the back lip **208***a* of the transfer ramp **102** is positioned above the top platform **122** of the liftgate **120** or above the top surface **232** of the other structure **230**, and/or after the front lip **208***b* of the transfer ramp **102** clears a front face **329** of the liftgate **120** (e.g., a front face of the first and second sides **126***a* and **126***b* and/or of the top platform **122**), the cable release latch **111** can be actuated via the actuation mechanism **115** and the cable **112** of the ramp latching system **110** to release a latchable portion **309** of the transfer ramp **102**. In turn, the transfer ramp **102** can be pivoted forward in a direction generally along or parallel to arrow B, such as (i) freely or (ii) via use of one or more springs (not shown) and/or by extending the struts **207***a* (FIG. **2**) and **207***b*. In some embodiments, the transfer ramp **102** can be rotated forward until the top surface **102***a* and/or the bottom surface **102***b* of the transfer ramp **102** is generally parallel to the top platform **122** of the liftgate **120**. In these and other embodiments, the transfer ramp **102** can be rotated forward until the back lip **208***a* of the transfer ramp **102** is positioned over the top surface **232** of the other structure **230** and/or until the front lip **208***b* of the transfer ramp **102** is positioned over the top platform **122** of the liftgate **120**.

(56) In FIG. **3**D, with the transfer ramp **102** rotated forward, the extendable arms **104***a* and **104***b* can be retracted to lower the transfer ramp **102** in a direction generally parallel to arrow C and toward the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**. As shown in FIG. **3**E, the extendable arms **104***a* and **104***b* can be retracted until the bottom surface **102***b* of the transfer ramp **102** abuts or otherwise rests on the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**. As discussed above, in the deployed state shown, the transfer ramp **102** is in alignment with the top platform **122** of the liftgate **120**, is in contact with the top platform **122** of the liftgate **120**, is in contact with the top surface **232** of the other structure **230**, and bridges the gap between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**.

(57) In the illustrated embodiment, the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230** are positioned at relatively the same height. Thus, as the transfer ramp **102** is lowered to the deployed state shown in FIG. **3**E, the transfer ramp **102** bridges the gap while remaining relatively flat, horizontal, or level. In scenarios in which the top platform **122** of the

liftgate **120** is positioned at a different height from the top surface **232** of the other structure **230**, the transfer ramp **102** can settle into a deployed state in which the transfer ramp **102** (*a*) rests on the top platform **122**, (b) rests on the top surface **232**, (c) bridges the gap between the top platform **122** and the top surface **232**, and (d) is positioned in a slanted or ramped orientation.
(58) In some embodiments, the extendable arms **104** can be retracted beyond the point at which the transfer ramp **102** initially contacts the top platform **122** and the top surface **232**. For example, continued retraction of the extendable arms **104** can press or smash the transfer ramp **102** into the top platform **122** and the top surface **232** to bring the top platform **122** and the top surface **232** into closer alignment with one another and/or to bring the transfer ramp **102** closer to a relatively flat, horizonal, or level orientation. As a specific example, consider a scenario in which the top platform **122** is positioned at a lower height than the top surface **232**. In this scenario, continued retraction of the extendable arms **104** beyond the point at which the transfer ramp **102** initially contacts the top platform **122** and the top surface **232** can lower the height of the top surface **232** (e.g., via the suspension of a vehicle that includes the top surface **232**) and/or can raise the height of the top platform **122** (e.g., via an upwards force that results as the transfer ramp **102** is pushed down into the top surface **232**). As a result, the height of the top platform **122** can be brought into alignment with the height of the top surface **232**, meaning that the transfer ramp **102** can be brought to (or closer to) a generally flat, horizontal, or level orientation thereby making transfers of cargo across the transfer ramp **102** easier.
(59) In these and other embodiments, all or a portion of the transfer ramp **102** can be formed of a material (e.g., aluminum or another suitable material) that is conformable and/or resiliently deformable. In these embodiments, (a) when the top platform **122** of the liftgate **120** is positioned at a different height from the top surface **232** of the other structure **230** and/or (b) when the top platform **122** and/or the top surface **232** are not level or flat relative to the transfer ramp **102** such that one side of the transfer ramp **102** (at or near one of the sidewalls **103***a* or **103***b*) makes contact with the top platform **122** and/or the top surface **232** before the other side of the transfer ramp **102** (at or near the other of the sidewalls **103***a* and **103***b*), at least one of the extendable arms **104** can continue to be retracted (e.g., generally along or parallel to arrow D shown in FIG. **3**E) after a portion of the transfer ramp **102** initially makes contact with the top platform **122** and/or the top surface **232**. Continuing to retract one or more of the extendable arms **104** in this manner can facilitate conforming the transfer ramp **102** to the top platform **122** and/or the top surface **232** to provide a suitably flat surface for transferring cargo between the top platform **122** and the top surface **232** across the transfer ramp **102**.
(60) As a specific example, consider a scenario in which a first side of the top surface **232** is positioned higher (closer) to the transfer ramp **102** in the generally flat or horizontal orientation than a second side of the top surface **232** opposite the first side of the top surface **232**. In this example, as the extendable arms **104** are retracted to lower the transfer ramp **102** toward the top surface **232**, a first side of the transfer ramp **102** (e.g., a side of the transfer ramp **102** that is connected to the first extendable arm **104***a*) can make initial contact with the first side of the top surface **232** before a second side of the transfer ramp **102** (e.g., a side of the transfer ramp **102** that is connected to the second extendable arm **104***b*) makes contact with the second side of the top surface **232**. When the first side of the transfer ramp **102** makes initial contact with the first side of the top surface **232**, the second side of the transfer ramp **102** may not be in contact with the second side of the top surface **232**, and may instead be positioned a distance above the second side of the top surface **232**. As such, at this point in time, the second side of the transfer ramp **102** can pose a roadblock or obstacle to transferring cargo across at least the second side of the transfer ramp **102**. Thus, after the first side of the transfer ramp **102** initially makes contact with the first side of the top surface **232**, one or more of the extendable arms **104** can continue to be retracted to lower the second side of the transfer ramp **102**. In embodiments employing a respective actuator for each of the first and second extendable arms **104***a* and **104***b*, continued retraction of the extendable arms

104 can overcurrent the actuator controlling the first extendable arm **104**_a_ (e.g., due to the first side of the transfer ramp **102** already being in contact with the first side of the top surface **232**) such that the actuator controlling the first extendable arm **104**_a_ ceases to retract the first extendable arm **104**_a_. At the same time, the actuator controlling the second extendable arm **104** can continue to retract the second extendable arm **104** to pull the second side of the transfer ramp **102** toward the second side of the top surface **232**. In turn, the transfer ramp **102** can bend (e.g., deform, warp) such that the second side of the transfer ramp **102** is brought into contact with the second side of the top surface **232** while the first side of the transfer ramp **102** remains in contact with the first side of the top surface **232**. In other words, the continued retraction of at least the second extendable arm **104**_b_ can pull the second side of the transfer ramp **102** against the second side of the top surface **232** such that the transfer ramp **102** is conformed to the top surface **232** and/or the top platform **122**, despite the initial misalignment. Conforming the transfer ramp **102** to the top platform **122** and/or the top surface **232** in this manner can reduce or eliminate the roadblock/obstacle posed by the second side of the transfer ramp **102**, and/or can provide a more suitable or flat surface for transferring cargo across the transfer ramp **102** between the top platform **122** and the top surface **232**. In embodiments in which the transfer ramp **102** is resiliently deformable, the transfer ramp **102** can return to its initial, unbent, and/or non-conforming shape as the extendable arms **104** are subsequently extended to raise the transfer ramp **102** off of the top surface **232** and/or the top platform **122**.

(61) Additionally, or alternatively, when the first side of the transfer ramp **102** makes initial contact with the first side of the top surface **232** in the above example and the second side of the transfer ramp **102** is not in contact with the second side of the top surface **232**, both of the extendable arms **104** can continue to be retracted. In some embodiments, this can force the first side of the transfer ramp **102** down into the first side of the top surface **232** to lower the height of the first side of the top surface **232** relative to the second side of the top surface **232** (e.g., using the suspension of a vehicle including the top surface **232**). Lowering the height of the first side of the top surface **232** relative to the second side of the top surface **232** can bring the second side of the transfer ramp **102** into contact with the second side of the top surface **232**, thereby (a) leveling the transfer ramp **102** and/or the top surface **232** and/or (b) eliminating, reducing, or minimizing the roadblock or obstacle posed by the second side of the transfer ramp **102** to transferring cargo across at least the second side of the transfer ramp **102**.

(62) FIGS. **4**A and **4**B are partially schematic perspective views of the first and second extendable arms **104**_a_ and **104**_b_, respectively, of the system **100** of FIG. **1**. As shown, the first and second extendable arms **104**_a_ and **104**_b_ can each include a rail or sliding system **440**_a_ and **440**_b_, respectively. Referring to the rail system **440**_a_ of the first extendable arm **104**_a_ of FIG. **4**A as an example, the rail system **440**_a_ includes first through fifth rail portions **441**_a_-**445**_a_. The first rail portion **441**_a_ is fixedly attached to the first mounting bracket **106**_a_. The second rail portion **442**_a_ is illustrated as an ‘I’ beam having a channel on each of its longitudinal sides. The first rail portion **441**_a_ is seated in a channel on a first longitudinal side of the second rail portion **442**_a_, and the third rail portion **443**_a_ is seated in a channel on a second longitudinal side of the second rail portion **442**_a_. The third rail portion **443**_a_ is illustrated as having a ‘C’ channel. The fourth rail portion **444**_a_ is positioned within the ‘C’ channel of the third rail portion **443**_a_, and is fixedly attached to the fifth rail portion **445**_a_. The fifth rail portion **445**_a_ is attached to the first side rail **203**_a_ of the transfer ramp **102**. In particular, the fifth rail portion **445**_a_ is attached to the first side rail **203**_a_ at a point **449**_a_ (e.g., a hinge pin) and in a manner that permits the transfer ramp **102** to pivot about the point **449**_a_ (e.g., as the strut **207**_a_ is extended and/or retracted). As shown in FIG. **4**B, the rail system **440**_b_ of the second extendable arm **104**_b_ is structured in a manner generally similar to the rail system **440**_a_ (FIG. **4**A) of the first extendable arm **104**_a_ (FIG. **4**A), but is fixedly attached to the second mounting bracket **106**_b_ and is pivotally attached to the second side rail **203**_b_ of the transfer ramp **102** at a point **449**_b_ (e.g., a hinge pin) and in a manner that permits the transfer ramp **102** to

pivot about the point **449***b* (e.g., as the strut **207***b* is extended and/or retracted).

(63) In the illustrated embodiment, each of the rail systems **440***a* and **440***b* are generally telescoping. The telescoping feature of the rail systems **440***a* and **440***b* facilitates deploying (e.g., raising) and retracting (e.g., lowering) the transfer ramp **102**. As discussed in greater detail below, the rail systems **440***a* and **440***b* can, via one or more actuators, be operated in unison (e.g., such that the rail systems **440***a* and **440***b* can be extended or retracted together and/or at the same time) or independently of one another.

(64) FIG. **5** is a partially schematic perspective view of an actuator **550** of the system **100** of FIG. **1**. As shown, the actuator **550** is positioned beneath the top platform **122** of the liftgate **120** and is attached to the second mounting bracket **106***b*. The actuator **550** is further attached to the fifth rail portion **445***b* of the rail system **440***b* of the second extendable arm **104***b* at location **547**. In other embodiments, the actuator **550** can be attached to the fifth rail portion **445***b* of the rail system **440***b* at another location different from the location **547**, and/or can be attached to another component of the second extendable arm **104***b* besides the fifth rail portion **445***b*.

(65) In the illustrated embodiment, the actuator **550** is a linear actuator that is configured to extend and retract. As the actuator **550** extends, the actuator **550** pushes on the fifth rail portion **445***b* of the rail system **440***b*, thereby causing the telescoping rail system **440***b* of the second extendable arm **104***b* to extend outward and raise the transfer ramp **102**. As the actuator **550** retracts, the actuator **550** pulls the fifth rail portion **445***b* of the rail system **440***b* inward, thereby causing the telescoping rail system **440***b* to retract inward and lower the transfer ramp **102**. In embodiments employing a single (e.g., only one) actuator **550** or only one or more actuators **550** coupled to the second extendable arm **104***b* (as opposed to the first extendable arm **104***a*), the first extendable arm **104***a* can be extended and retracted via extension and retraction, respectively, of the second extendable arm **104***b* when the actuator **550** is extended and retracted, respectively.

(66) The actuator **550** can be a pneumatic, hydraulic, electric (e.g., electro-hydraulic, electromechanical), and/or piezoelectric linear actuator. In embodiments employing a pneumatic linear actuator, the linear actuator can be tied into an air system of the vehicle/trailer. For example, the pneumatic linear actuator can be tapped into an air suspension system of the vehicle/trailer (e.g., as an accessory). Additionally, or alternatively, using a pneumatic linear actuator can facilitate controlled stowage of the transfer ramp **102** and/or of the system **100** by venting air pressure (e.g., through a valve). In other embodiments, the actuator **550** can be another suitable type of actuator system, such as a chain and pulley system. Operation of the actuator **550** can be controlled via one or more user controls (e.g., push buttons, levers, toggle switches, etc.), such as user controls mounted or stowed on the liftgate **120**, a side of the vehicle/trailer, or within a cabin or box of the vehicle. In some embodiments, the user controls for the actuator **550** can be positioned proximate the controls for the liftgate **120**.

(67) Although shown as being coupled to the second extendable arm **104***b* in FIG. **5**, the actuator **550** can be coupled to the first extendable arm **104***a* in other embodiments of the present technology. In some embodiments, the system **100** includes an actuator **550** attached to the first extendable arm **104***a* in addition to the actuator **550** attached to the second extendable arm **104***b*. In these embodiments, the actuator **550** attached to the first extendable arm **104***a* and the actuator **550** attached to the second extendable arm **104***b* can extend and/or retract in unison such that the first and second extendable arms **104***a* and **104***b* are extended and/or retracted, respectively, in unison. In these and other embodiments, the actuator **550** attached to the first extendable arm **104***a* can be extended and/or retracted independently from the actuator **550** attached to the second extendable arm **104***b* (and/or vice versa).

(68) FIGS. **6**A-**6**C are partially schematic perspective views of the ramp latching system **110** of the system **100** of FIG. **1**. Referring to FIGS. **6**A and **6**B, the latchable portion **309** of the transfer ramp **102** is shown latched in the cable release latch **111** of the ramp latching system **110**. When the latchable portion **309** of the transfer ramp **102** is latched in the cable release latch **111**, the transfer

ramp **102** can be held in the stowed orientation (shown in FIG. **1**) and prevented or hindered from being rotated or pivoted forward toward the generally flat or horizontal orientation (shown in FIG. **2**). Referring now to FIG. **6**C, when a user of the system actuates the cable release latch **111** via the actuation mechanism **115** and the cable **112**, the latchable portion **309** of the transfer ramp **102** is released from the cable release latch **111** such that the transfer ramp **102** can be rotated or pivoted from the stowed orientation toward the generally flat or horizontal orientation. In some embodiments, the ramp latching system **110** can operate in a manner generally similar to a hood release latch in an automobile. In these and other embodiments, another suitable type of latching system can be employed in lieu of the ramp latching system **110** and/or the latchable portion **309** of the transfer ramp **102**. Examples of other suitable types of latching systems include cotter pins, ball lock pins, push button locks, and thumb latches. Several of these other suitable types of latching systems are discussed in greater detail below.

(69) In some embodiments, the transfer ramp **102** can be configured to freely rotate (e.g., without biasing) when the latchable portion **309** of the transfer ramp **102** is released from the cable release latch **111**. In other embodiments, the system **100** can include one or more springs (not shown) that are useable (in addition to or in lieu of the struts **207***a* and/or **207***b*) to bias, transition, or pivot the transfer ramp **102** at least part of the way between the stowed orientation (shown in FIG. **1**) and the generally flat or horizontal orientation (shown in FIG. **2**). For example, the system **100** can include one or more springs (e.g., one or more torsion springs and/or one or more other suitable types of springs) that are installed on, at, or proximate the point **449***a* (FIG. **4**A) and/or the point **449***b* (FIG. **4**B). Continuing with this example, when the latchable portion **309** of the transfer ramp **102** is released from the cable release latch **111**, the one or more springs can bias or force the transfer ramp **102** toward the generally flat or horizontal orientation. As a specific example, when the transfer ramp **102** is transitioned to the stowed orientation (shown in FIG. **1**), the one or more springs can be twisted (e.g., wound) such that the one or more springs are transitioned from storing a first amount of energy to storing a second, higher amount of energy. Thus, when the transfer ramp **102** is released from the cable release latch **111**, the one or more springs are permitted to untwist (e.g., unwind) and release the stored. As this occurs, the one or more springs can exert a torque or twisting force against the transfer ramp **102** to transition, pivot, or rotate the transfer ramp **102** at least part of the way between the stowed orientation and the generally flat or horizontal orientation.

(70) FIGS. **7**A and **7**B are partially schematic perspective views of the strut **207***b* of the system **100** of FIG. **1**. The strut **207***b* can be fixedly attached to the second side rail **203***b* of the transfer ramp **102** and/or to the second extendable arm **104***b*. The strut **207***a* (FIG. **2**) can be structurally generally similar to the strut **207***b*, but can be fixedly attached to the first side rail **203***a* of the transfer ramp **102** and/or to the first extendable arm **104***a*.

(71) In some embodiments, the struts **207***a* and **207***b* can be gas struts. In other embodiments, the struts **207***a* and **207***b* can be air cylinders. In still other embodiments, the struts **207***a* and **207***b* can be other suitable actuator systems, such as a pneumatic, hydraulic, electric (e.g., electro-hydraulic, electromechanical), and/or piezoelectric linear actuators or pistons. In still other embodiments, the struts **207***a* and **207***b* can be controlled via stepper motors or other actuators. In embodiments in which the struts **207***a* and **207***b* are gas struts or air cylinders, the struts **207***a* and **207***b* can be tapped into an air suspension system of the vehicle/trailer (e.g., as accessories).

(72) As discussed above, the struts **207***a* and **207***b* can be used to rotate or pivot the transfer ramp **102**. For example, after the transfer ramp **102** is released from the cable release latch **111** of the ramp latching system **110** and/or after the spring(s) (not shown) pivot the transfer ramp **102** at least part of the way toward the generally flat or horizontal orientation (shown in FIG. **2**), the struts **207***a* and **207***b* can be extended, such as from the position of the strut **207***b* shown in FIG. **7**A to the position of the strut **207***b* shown in FIG. **7**B. In embodiments in which the struts **207***a* and **207***b* are gas struts or air cylinders, the struts **207***a* and **207***b* can be extended by filling the strut **207***a* and/or the strut **207***b* with gas or air. Extension of the struts **207***a* and **207***b* can rotate or pivot the transfer

ramp **102** away from the stowed orientation (shown in FIG. **1**) and/or toward the flat or horizontal orientation (shown in FIG. **2**), or further.

(73) Additionally, or alternatively, the struts **207***a* and **207***b* can be retracted, such as from the position of the strut **207***b* shown in FIG. **7**B to the position of the strut **207***b* shown in FIG. **7**A. For example, in embodiments in which the struts **207***a* and **207***b* are gas struts or air cylinders, gas or air in the strut **207***a* and/or the strut **207***b* can be vented via a valve (not shown) such that the struts **207***a* and **207***b* are permitted to retract. Additionally, or alternatively, a user of the system **100** can (a) pull on a rope or cable (not shown) attached to (e.g., a back portion) of the transfer ramp **102** and/or (b) push on (e.g., a back portion of the top surface **102***a* of) the transfer ramp **102**, to retract the struts **207***a* and **207***b* and/or rotate or pivot the transfer ramp **102** toward the stowed orientation. In some embodiments, retraction of the struts **207***a* and **207***b* can rotate or pivot the transfer ramp **102** from the flat or horizontal orientation toward the stowed orientation. In these and other embodiments, the system **100** can include a transfer ramp stowage system (e.g., similar to the transfer ramp stowage system discussed in detail below with reference to FIGS. **25**A-**25**D and/or similar to the transfer ramp stowage system discussed in detail below with reference to FIGS. **26**A-**26**E). For example, the system **100** can include a cable (not shown) connected to a portion of the transfer ramp **102**. The cable can be routed from the portion of the transfer ramp **102**, across a pulley or other object (e.g., through a loop), and/or to a position (e.g., a fixed position) on the system **100** such that, when the extendable arms **104** are extended while the latchable portion **309** of the transfer ramp **102** is released from the cable release latch **111**, the cable can be pulled taught and used to pivot or rotate the transfer ramp **102** toward the stowed orientation (shown in FIG. **1**). More specifically, as the extendable arms **104** are extended, the cable connected to the transfer ramp **102** can be brought taught (e.g., against the pulley or other object) to apply a force against the transfer ramp **102** and rotate the transfer ramp **102** toward the stowed position. Such a force can additionally be used to retract the struts **207***a* and/or **207***b* in some embodiments. In these and other embodiments, as the transfer ramp **102** returns to the stowed orientation, the latchable portion **309** of the transfer ramp **102** can engage with the cable release latch **111** of the ramp latching system **110** such that the cable release latch **111** latches the latchable portion **309** of the transfer ramp **102** and prevents or hinders the transfer ramp **102** from rotating or pivoting away from the stowed orientation.

(74) In some embodiments, only one of the struts **207***a* and **207***b* are actively actuated (e.g., actively filled with gas or air and/or vented). In these embodiments, the other of the struts **207***a* and **207***b* can extend or retract via active extension or retraction of the one of the struts **207***a* and **207***b*. In embodiments in which both of the struts **207***a* and **207***b* are actively actuated, the struts **207***a* and **207***b* can be actively actuated such that the struts **207***a* and **207***b* extend and/or retract in unison. In other embodiments, the struts **207***a* and **207***b* can be actively actuated such that the struts **207***a* and **207***b* can extend and/or retract independent of one another.

(75) In some embodiments, the struts **207***a* and **207***b* can limit over rotation or pivoting of the transfer ramp **102**. For example, the struts **207***a* and **207***b* can be extended out to a certain distance or extent beyond which further extension can be prevented or hindered. In this manner, the struts **207***a* and **207***b* can limit rotation or pivoting of the transfer ramp **102** to positions corresponding to extension of the struts **207***a* and **207***b* beyond the certain distance or extent. As another example, the struts **207***a* and **207***b*, the cable release latch **111**, and/or the latchable portion **309** of the transfer ramp **102** can limit over rotation or pivoting of the transfer ramp **102** beyond the stowed orientation of the transfer ramp **102** (as shown in FIG. **1**).

(76) FIGS. **8**A-**8**L are partially schematic side views of the transfer ramp system **100** of FIG. **1** illustrating a method of operating the system **100** in accordance with various embodiments of the present technology. All or a subset of the steps of the method illustrated in FIGS. **8**A-**8**L can be performed in accordance with the above discussion of the system **100**. As shown in FIG. **8**A, the method begins with the system **100** in the stowed position beneath the top platform **122** of the

liftgate **120**. More specifically, the vehicle/trailer can initially be backed toward the other structure **230** until (a) the vehicle/trailer is positioned such that the rubber bumper **127** of the liftgate **120** faces and/or contacts the other structure **230**, and/or (b) the top platform **122** of the liftgate **120** is spaced a distance from the top surface **232** of the other structure **230** that is less than a length of the transfer ramp **102** measured from the back lip **208***a* (FIG. **2**) of the transfer ramp **102** (FIG. **1**) to the front lip **208***b* (FIG. **2**) of the transfer ramp **102**. As discussed above, the other structure **230** can be a commercial vehicle, a loading dock, a trailer, a bed of a truck, or another structure. When the vehicle/trailer including the liftgate **120** is positioned as shown in FIG. **8**A, a gap G exists between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230**.

(77) Referring now to FIGS. **8**B-**8**D, the method continues by raising the transfer ramp **102** from the stowed position (FIG. **8**A) to a deployed position (FIG. **8**D) above the top platform **122** of the liftgate **120** and/or above the top surface **232** of the other structure **230**. Raising the transfer ramp **102** can include extending the first and second extendable arms **104***a* (FIG. **1**) and **104***b* using the actuator **550** and/or the rail systems **440***a* and **440***b*. Raising the transfer ramp **102** using the actuator **550** can include filling the actuator **550** with gas or air, or otherwise actuating the actuator **550**. In some embodiments, raising the transfer ramp **102** includes raising the transfer ramp **102** such that the transfer ramp **102** passes through the gap G that exists between the top surface **232** of the other structure **230** and the top platform **122** of the liftgate **120**. In these and other embodiments, raising the transfer ramp **102** includes raising the transfer ramp **102** such that at least a portion of the transfer ramp **102** overshoots the top surface **232** of the other structure **230** (as shown in FIG. **8**D). In these and still other embodiments, raising the transfer ramp **102** includes raising the transfer ramp **102** until (i) the back lip **208***a* and/or a portion of the bottom surface **102***b* of the transfer ramp **102** is positioned above the top surface **232** of the other structure **230**, and/or (b) the front lip **208***b* (FIG. **2**) of the transfer ramp **102** clears (e.g., is positioned in front of) the front face **329** of the liftgate **120**.

(78) Referring to FIG. **8**E, the method continues by rotating or pivoting the transfer ramp **102** from the stowed orientation (FIG. **8**D) to a deployed orientation (e.g., in which the transfer ramp **102** is relatively flat or horizontal, and/or in which to top surface **102***a* and/or the bottom surface **102***b* of the transfer ramp **102** is/are brought more into alignment with the top platform **122** of the liftgate **120** and/or the top surface **232** of the other structure **230**). In some embodiments, rotating or pivoting the transfer ramp **102** can include releasing the latchable portion **309** of the transfer ramp **102** from the cable release latch **111** of the ramp latching system **110**, such as by actuating the cable release latch **111** via the actuation mechanism **115** and the cable **112** (FIG. **1**). In these and other embodiments, rotating or pivoting the transfer ramp **102** can include rotating or pivoting the transfer ramp **102** using one or more springs and/or by extending the struts **207***a* (FIG. **2**) and **207***b* (e.g., such that the transfer ramp **102** pivots about the points **449***a* (FIG. **4**A) and **449***b* at which the first and second extendable arms **104***a* (FIG. **1**) and **104***b*, respectively, are attached to the first and second side rails **203***a* (FIG. **2**) and **203***b*, respectively, of the transfer ramp **102**). Extending the struts **207***a* and **207***b* can include filling the struts **207***a* and/or **207***b* with gas or air, or otherwise actuating the struts **207***a* and/or **207***b*. In these and other embodiments, extending the struts **207***a* and **207***b* can include pushing on the transfer ramp **102** or otherwise manually rotating or pivoting the transfer ramp **102**.

(79) Referring to FIGS. **8**F and **8**G, the method continues by lowering the transfer ramp **102** until the transfer ramp **102** is in the deployed position illustrated in FIG. **8**G. Lowering the transfer ramp **102** can include lowering the transfer ramp **102** while the transfer ramp **102** is in the generally flat or horizontal orientation (or in another orientation that is more in alignment with the top platform **122** of the liftgate **120** than the stowed orientation of the transfer ramp **102**). Lowering the transfer ramp **102** can additionally, or alternatively, include retracting the first and second extendable arms **104***a* (FIG. **1**) and **104***b* using the actuator **550** and/or the rail systems **440***a* and **440***b*. Lowering the transfer ramp **102** using the actuator **550** can include venting gas or air from the actuator **550** (e.g.,

through a valve), or otherwise actuating the actuator **550**. In some embodiments, lowering the transfer ramp **102** includes lowering the transfer ramp **102** such that at least a portion of the bottom surface **102***b* of the transfer ramp **102** is positioned above the top surface **232** of the other structure **230** and at least another portion of the bottom surface **102***b* is positioned above the top platform **122** of the liftgate **120**. In these and other embodiments, lowering the transfer ramp **102** can include lowering the transfer ramp **102** until a portion of the bottom surface **102***b* of the transfer ramp **102** rests on the top surface **232** of the other structure **230**, another portion of the bottom surface **102***b* rests on the top platform **122** of the liftgate **120**, and the transfer ramp **102** spans the gap G. In these and still other embodiments, lowering the transfer ramp **102** can include continuing to retract at least one of the extendable arms **104** after the transfer ramp **102** makes initial contact with the top platform **122** and/or the top surface **232** (e.g., such that the transfer ramp **102** bends and/or conforms to the arrangement of the top platform **122** and/or the top surface **232**, as discussed in greater detail above).
(80) With the transfer ramp **102** positioned as shown in FIG. **8**G, the method continues by transferring cargo across the top surface **102***a* (FIG. **2**) of the transfer ramp **102**. For example, cargo (e.g., a pallet or other object) can be loaded into the vehicle/trailer by transferring the cargo from the top surface **232** of the other structure **230**, across the top surface **102***a* of the transfer ramp **102**, and onto the top platform **122** of the liftgate **120**. Additionally, or alternatively, cargo can be unloaded from the vehicle/trailer by transferring the cargo from the top platform **122** of the liftgate **120**, across the top surface **102***a* of the transfer ramp **102**, and onto the top surface **232** of the other structure **230**.
(81) Referring to FIG. **8**H, the method continues by raising the transfer ramp **102** from the deployed position (shown in FIG. **8**G) to a deployed position (shown in FIG. **8**H) above the top platform **122** of the liftgate **120** and/or above the top surface **232** of the other structure **230**. The deployed position shown in FIG. **8**H can be identical or at least generally similar to the deployed position shown in FIG. **8**E. Raising the transfer ramp **102** can include raising the transfer ramp **102** while the transfer ramp **102** is in the generally flat or horizontal orientation (or in another orientation that is more in alignment with the top platform **122** of the liftgate **120** than the stowed orientation of the transfer ramp **102**). Raising the transfer ramp **102** can additionally, or alternatively, include extending the first and second extendable arms **104***a* (FIG. **1**) and **104***b* using the actuator **550** and/or the rail systems **440***a* and **440***b*. Raising the transfer ramp **102** using the actuator **550** can include filling the actuator **550** with gas or air, or otherwise actuating the actuator **550**. In some embodiments, raising the transfer ramp **102** can include raising the transfer ramp **102** to a position at which the bottom surface **102***b* of the transfer ramp **102** is not in contact with the top surface **232** of the other structure **230** and/or with the top platform **122** of the liftgate **120**. In these and other embodiments, raising the transfer ramp **102** can include raising the transfer ramp **102** until the front lip **208***b* of the transfer ramp **102** clears or is positioned in front of the front face **329** of the liftgate **120**.
(82) Referring now to FIGS. **81** and **8**J, the method continues by rotating or pivoting the transfer ramp **102** to the stowed orientation. In some embodiments, rotating or pivoting the transfer ramp **102** to the stowed orientation can include retracting the struts **207***a* (FIG. **2**) and **207***b* (e.g., such that the transfer ramp **102** pivots about the points **449***a* (FIG. **4**A) and **449***b* at which the first and second extendable arms **104***a* (FIG. **1**) and **104***b*, respectively, are attached to the first and second side rails **203***a* (FIG. **2**) and **203***b*, respectively, of the transfer ramp **102**). Retracting the struts **207***a* and **207***b* can include venting gas or air from the struts **207***a* and/or **207***b* (e.g., through a valve), or otherwise actuating the struts **207***a* and/or **207***b*. In these and other embodiments, retracting the struts **207***a* and **207***b* can include pushing on (e.g., the top surface **102***a* (FIG. **2**) of) the transfer ramp **102**, pulling a cable or strap attached to the transfer ramp **102**, or otherwise manually rotating or pivoting the transfer ramp **102**. In these and still other embodiments, rotating or pivoting the transfer ramp **102** to the stowed orientation can include rotating or pivoting the transfer ramp **102**

using a transfer ramp stowage system (as discussed above and in greater detail below with reference to FIGS. **25**A-**25**D and/or FIGS. **26**A-**26**E), such as by extending the extendable arms **104** to pull a cable (not shown) connected to the transfer ramp **102** taught and thereby rotate the transfer ramp **102** toward the stowed orientation. In these and other embodiments, rotating or pivoting the transfer ramp **102** to the stowed orientation can include rotating or pivoting the transfer ramp **102** until the latchable portion **309** (FIG. **8**E) engages with and becomes latched by the cable release latch **111** of the ramp latching system **110** (e.g., such that rotation or pivoting away from the stowed orientation is prevented or hindered by the cable release latch **111**).

(83) Referring to FIGS. **8**K and **8**L, the method continues by lowering the transfer ramp **102** and returning the system **100** to the stowed position. Lowering the transfer ramp **102** can include lowering the transfer ramp **102** while the transfer ramp **102** is in the stowed orientation. Lowering the transfer ramp **102** can additionally, or alternatively, include retracting the first and second extendable arms **104***a* (FIG. **1**) and **104***b* using the actuator **550** and/or the rail systems **440***a* and **440***b*. Lowering the transfer ramp **102** using the actuator **550** can include venting gas or air from the actuator **550** (e.g., through a valve), or otherwise actuating the actuator **550**. In some embodiments, lowering the transfer ramp **102** includes lowering the transfer ramp **102** such that the transfer ramp **102** passes through the gap G that exists between the top surface **232** of the other structure **230** and the top platform **122** of the liftgate **120**. In these and other embodiments, lowering the transfer ramp **102** includes lowering the transfer ramp **102** until the system **100** is in the stowed position beneath the top platform **122** of the liftgate **120**.

(84) Although the steps of the methods described herein are discussed and/or illustrated in a particular order, the methods described herein are not so limited. In other embodiments, the methods described herein can be performed in different orders. In these and other embodiments, any of the steps of the methods described herein can be performed before, during, and/or after any of the other steps of the methods described herein. Furthermore, a person skilled in the art will readily recognize that the methods described herein can be altered and still remain within these and other embodiments of the present technology. For example, one or more steps of the methods described herein can be omitted and/or repeated in some embodiments.

(85) Variations of the embodiments illustrated in FIGS. **1**-**8**L and described in detail above exist and are within the scope of the present technology. Indeed, variations of the embodiments illustrated in FIGS. **1**-**8**L are discussed in detail below with reference to FIGS. **18**A-**27**. In these and other variations, transfer ramps **102** configured in accordance with other embodiments of the present technology can lack side rails **203***a* and/or **203***b*, and/or can include side rails that extend downward toward the bottom surfaces **102***b* of the transfer ramps **102** (as opposed to extending upward toward the top surfaces **102***a* of the transfer ramps **102**). As another example, the first and/or second extendable arms **104***a* and/or **104***b* can be attached to transfer ramps **102** at locations along the transfer ramps **102** that differ from the points **449***a* and/or **449***b* shown in FIGS. **4**A and/or **4**B, respectively. Additionally, or alternatively, the struts **207***a* and/or **207***b* (a) can be attached to transfer ramps **102** at locations along the transfer ramps **102** that differ from the locations shown in FIGS. **4**A and/or **7**A, respectively, and/or (b) can be attached to the first and/or second extendable arms **104***a* and/or **104***b*, respectively, at other locations along the first and/or second extendable arms **104***a* and/or **104***b* that differ from the locations shown in FIGS. **4**A and/or **4**B, respectively. As still another example, the first and/or second extendable arms **104***a* and/or **104***b* can include other extendable and/or retracting structures than shown in FIGS. **4**A and/or **4**B, respectively. In still other embodiments, the system **100** can include a different number of (e.g., one or more than two) extendable arms **104**.

(86) In these and other embodiments, the system **100** can be operated such that the transfer ramp **102** does not rest on the top platform **122** of the liftgate **120** and/or on the top surface **232** of the other structure **230** in the deployed position. For example, the transfer ramp **102** can be deployed such that the transfer ramp **102**, in the deployed position, rests on the top platform **122** of the

liftgate **120** or the top surface **232** of the other structure **230**, but not both. As another example, the transfer ramp **102** can be deployed such that the transfer ramp **102** bridges the gap G that exists between the top platform **122** of the liftgate **120** and the top surface **232** of the other structure **230** by fitting within the gap G without resting on the top platform **122** of the liftgate **120** and/or without resting on the top surface **232** of the other structure **230**.

(87) In some embodiments, the system **100** can additionally include a spring that biases the transfer ramp **102** toward the stowed orientation. The spring can be used in addition to or in lieu of the struts **207***a* and/or **207***b*. The transfer ramp **102** in some of these embodiments can be manually pivoted or rotated from the stowed orientation toward the generally flat or horizontal orientation, such as stepping on, pushing on, pulling on, or otherwise manually tilting the transfer ramp **102** toward the generally flat or horizontal orientation. Alternatively, the transfer ramp **102** can be rotated or tilted toward the flat or horizontal orientation, for example, by extending the struts **207***a* and/or **207***b* and/or using another actuator. Thereafter, the spring can be used to return the transfer ramp **102** back to the stowed orientation.

(88) Furthermore, although many of the steps of the methods described above are discussed as involving users of the system **100**, one or more of these steps can be automated in some embodiments of the present technology. For example, the system **100** can be installed on a fully autonomous vehicle or trailer towed by a fully autonomous vehicle. In these and other embodiments, the process of backing up the liftgate **120** toward the other structure **230** (e.g., before deploying the transfer ramp **102**) can be autonomous. In these and still other embodiments, extending and/or retracting the first and/or second extendable arms **104***a* and/or **104***b* to raise and/or lower, respectively, the transfer ramp **102** can be autonomous. Additionally, or alternatively, releasing the transfer ramp **102** from the ramp latching system **110** (e.g., by actuating the cable release latch **111**) can be autonomous; rotating and/or pivoting the transfer ramp **102** (e.g., by extending and/or retracting the struts **207***a* and/or **207***b*) can be autonomous; and/or transferring cargo across the top surface **102***a* of the transfer ramp **102** can be autonomous.

(89) FIGS. **9**A-**9**C are partially schematic side views of another transfer ramp system **900** ("the system **900**") configured in accordance with various embodiments of the present technology. As shown, the system **900** includes a transfer ramp **902** and at least one extendable arm **904**. The transfer ramp **902** and/or the extendable arm(s) **904** can be generally similar to the transfer ramp **102** (FIG. **1**) and the extendable arms **104** (FIG. **1**), respectively, of the system **100** (FIG. **1**) described above. For example, the extendable arm(s) **904** can be extended and/or retracted using an actuator (not shown), such as a linear actuator. In contrast with the system **100**, the system **900** is configured to pivot about multiple different axes. For example, the system **900** includes a first hinge or pivot point **961** and a second hinge or pivot point **962**.

(90) Referring to FIG. **9**A, the transfer ramp **902** can be advanced (raised) generally along or parallel to arrow A through a gap that exists between a top platform **922** (e.g., a dock plate) of a liftgate **920** and the top surface **232** of the other structure **230**. As shown in FIG. **9**B, after the transfer ramp **902** is raised through the gap, the transfer ramp **902** can be rotated or pivoted (e.g., using one or more struts) about the first pivot point **961** (generally along or parallel to arrow B) to move the transfer ramp **902** from a stowed orientation (FIG. **9**A) to a generally flat or horizontal orientation (FIG. **9**B). Thereafter, as shown in FIG. **9**C, the extendable arm(s) **904** can pivot about the second pivot point **962** and generally along or parallel to arrow C. Pivoting about the second pivot point **962** allows a bottom surface **902***b* of the transfer ramp **902** to fall against the top platform **922** of the liftgate **920** and the top surface **232** of the other structure **230**. In this manner, the transfer ramp **902** can be lowered (e.g., without needing to retract the extendable arm(s) **904**). Cargo can then be transferred across a top surface **902***a* of the transfer ramp **902**.

(91) To retract the system **900**, the extendable arm(s) **904** can pivot about the second pivot point **962** in a direction generally opposite to the arrow C shown in FIG. **9**C. Thereafter, the transfer ramp **902** can pivot or rotate about the first pivot point **961** in a direction generally opposite to the

arrow B shown in FIG. **9**B to return the transfer ramp **902** to the stowed orientation. Then, the transfer ramp **902** can be lowered using the extendable arm(s) **904** to return the system **900** to the retracted position or state beneath the liftgate **920**.

(92) FIGS. **10**A-**10**C are partially schematic cross-sectional side views of another transfer ramp system **1000** ("the system **1000**") configured in accordance with various embodiments of the present technology. The system **1000** includes a transfer ramp **1002** and a hinge or pivot point **1008**. In a stowed or undeployed state of the system **1000**, the transfer ramp **1002** can be positioned within a corresponding recess **1005** (FIGS. **10**B and **10**C) in a liftgate **1020**. To deploy the transfer ramp **1002**, the transfer ramp **1002** can be (a) lifted or raised out of the recess **1005** generally along or parallel to the arrow A (FIG. **10**A) and (b) rotated or pivoted about the pivot point **1008** generally along or parallel to the arrow B (FIG. **10**B) until (i) a bottom surface **1002***b* (FIGS. **10**B and **10**C) rests on the top surface **232** of the other structure **230** and (ii) the transfer ramp **1002** spans a gap between the top surface **232** and a top platform **1022** (e.g., a dock plate) of the liftgate **1020**. Thereafter, cargo can be transferred across a top surface **1002***a* (FIGS. **10**B and **10**C) of the transfer ramp **1002**. To retract or undeploy the system **1000**, the transfer ramp **1002** is (a) lifted or rotated about the pivot point **1008** in a direction generally opposite to the arrow B (FIG. **10**B) and (b) lowered back into the corresponding recess **1005** in the liftgate **1020**.

(93) FIGS. **11**A and **11**B are partially schematic side views of another transfer ramp system **1100** ("the system **1100**") configured in accordance with various embodiments of the present technology. As shown in FIG. **11**A, the system **1100** includes a transfer ramp **1102** that, in a stowed or undeployed state, serves as a top plate (e.g., a dock plate) of a liftgate **1120**. To deploy the transfer ramp **1102**, the transfer ramp **1102** is unfolded generally along or parallel to arrow A. Although shown with a single panel or section in FIG. **11**A, the transfer ramp **1102** can include multiple panels or sections in some embodiments. For example, referring to FIG. **11**B, the transfer ramp **1102** can include two panels or sections. The first and second sections of the transfer ramp **1102** can be unfolded from off a platform **1122** of the liftgate **1120** generally along or parallel to arrow A, and the second section of the transfer ramp **1102** can then be unfolded off of the first section of the transfer ramp **1102** generally along arrow B. When the transfer ramp **1102** of FIG. **11**A or **11**B is deployed, a bottom surface **1102***b* of the transfer ramp **1102** can rest on a top surface **232** of another structure **230** such that the transfer ramp **1102** bridges a gap that exists between the top surface **232** of the other structure **230** and the platform **1122** of the liftgate **1120**. Cargo can then be transferred across a top surface **1102***a* of the transfer ramp **1102**. To retract or undeploy the transfer ramp **1102**, the transfer ramp **1102** can be rotated or folded in a direction generally opposite to the arrow B (FIG. **11**B) and/or in a direction generally opposite to the arrow A (FIGS. **11**A and **11**B) such that sections of the transfer ramp **1102** are folded together and/or such that the transfer ramp **1102** is returned to the stowed position on top of the liftgate **1120** (as shown in FIG. **11**A).

(94) FIGS. **12**A-**12**D are partially schematic side views of another transfer ramp system **1200** ("the system **1200**") configured in accordance with various embodiments of the present technology. As shown, the system **1200** is generally similar to the system **1100** of FIGS. **11**A and **11**B in that the system **1200** includes a transfer ramp **1202** that rests on top of a liftgate **1220** and/or serves as a top plate (e.g., a dock plate) of the liftgate **1220** when the system **1200** is in a stowed or retracted position or state. The transfer ramp **1202** of the system **1200**, however, differs from the transfer ramp **1102** of the system **1100**. In particular, the transfer ramp **1202** includes multiple sections (identified individually in FIGS. **12**A-**12**D as first section **1202***y* and second section **1202***z*) that fold under one another. Although shown with two sections **1202***y* and **1202***z* in the illustrated embodiment, the transfer ramp **1202** can include a different number of sections (e.g., one or more than two sections) in other embodiments of the present technology.

(95) To deploy the system **1200**, the transfer ramp **1202** is initially lifted or raised such that the transfer ramp **1202** rotates about a hinge or pivot point **1261**, as shown in FIG. **12**B. Thereafter, as shown in FIG. **12**C, the second section **1202***z* of the transfer ramp **1202** is unfolded from under the

first section **1202***y* of the transfer ramp **1202** by rotating or pivoting the second section **1202***z* about a hinge or pivot point **1262** generally along or parallel to arrow A. With the second section **1202***z* of the transfer ramp **1202** unfolded, the transfer ramp **1202** is then lowered (as shown in FIG. **12**D) until a bottom surface **1202***b* of the second section **1202***z* comes to rest on a top surface of another structure (not shown) such that the transfer ramp **1202** spans a gap that exists between the liftgate **1220** and the other structure. Cargo can then be transferred across a top surface **1202***a* (FIG. **12**D) of the transfer ramp **1202**. To retract or undeploy the system **1200**, the transfer ramp **1202** can be raised, the second section **1202***z* can be rotated or pivoted about the pivot point **1262** in a direction generally opposite to the arrow A (FIG. **12**C) such that the second section **1202***z* is folded under the first section **1202***y* (as shown in FIG. **12**B), and the transfer ramp **1202** can be lowered until it rests on top of the liftgate **1220** (as shown in FIG. **12**A).

(96) FIGS. **13**A-**13**E are partially schematic perspective views of another transfer ramp system **1300** ("the system **1300**") configured in accordance with various embodiments of the present technology. The system **1300** includes a transfer ramp **1302** and tracks **1306** (identified individually in FIGS. **13**A-**13**E as first track **1306***a* and second track **1306***b*). The first and second tracks **1306***a* and **1306***b* are mounted to a liftgate **1320** beneath a top platform **1322** (e.g., a dock plate) of the liftgate **1320**. In the illustrated embodiment, the first and second tracks **1306***a* and **1306***b* are arced and define a non-linear pathway along which the transfer ramp **1302** can be slid or translated. The first and second tracks **1306***a* and **1306***b* can have another shape and/or define a different pathway for the transfer ramp **1302** in other embodiments of the present technology.

(97) In a stowed position or state of the system **1300**, the transfer ramp **1302** is positioned beneath the top platform **1322** of the liftgate **1320** and between the first and second tracks **1306***a* and **1306***b*. More specifically, the transfer ramp **1302** includes a first section **1302***y* and a second section **1302***z* that fold over one another. As best shown in FIG. **13**E, the first section **1302***y* is slidably attached to the first and second tracks **1306***a* and **1306***b* such that the transfer ramp **1302** can be moved along the first and second tracks **1306***a* and **1306***b* but is prevented from detaching from or coming out of the first and second tracks **1306***a* and **1306***b*. In some embodiments, the second section **1302***z* of the transfer ramp **1302** may rest within and/or be constrained by the first and second tracks **1306***a* and **1306***b* at least while the transfer ramp **1302** is stowed under the liftgate **1320**. Although shown with two sections in the illustrated embodiment, the transfer ramp **1302** can include a different number of (e.g., one or more than two) sections in other embodiments of the present technology.

(98) To deploy the system **1300**, the transfer ramp **1302** can initially be slid along the first and second tracks **1306***a* and **1306***b* in a direction generally along or parallel to arrow A (FIG. **13**B). As shown, the transfer ramp **1302** includes a handle **1305**. In these embodiments, a user of the system **1300** can pull on the handle **1305** to manually slide or translate the transfer ramp **1302** along the first and second tracks **1306***a* and **1306***b*. Additionally, or alternatively, movement of the transfer ramp **1302** can be automated. For example, the system **1300** can include one or more actuators for moving the transfer ramp **1302** along the first and second tracks **1306***a* and **1306***b*. In some of these embodiments, the handle **1305** can be omitted.

(99) As shown in FIG. **13**C, after moving the transfer ramp **1302** generally along or parallel to the arrow A (FIG. **13**B), the transfer ramp **1302** can then be slid vertically along the first and second tracks **1306***a* and **1306***b* in a direction generally along or parallel to arrow B. Again, a user can manually lift the transfer ramp **1302** vertically (e.g., using the handle **1305**), and/or the vertical movement of the transfer ramp **1302** can be automated via use of one or more actuators. Referring now to FIG. **13**D, the second section **1302***z* of the transfer ramp **1302** can be unfolded away from the first section **1302***y* of the transfer ramp **1302** by swinging the second section **1302***z* generally along or parallel to the arrow C.

(100) As shown in FIG. **13**E, after unfolding the transfer ramp **1302**, the transfer ramp **1302** can be moved to a generally flat or horizontal orientation by moving the transfer ramp **1302** generally

along or parallel to arrow D. More specifically, the first and second sections **1302***y* and **1302***z* can be brought in line with one another, and/or the transfer ramp **1302** can be moved such that a bottom surface **1302***b* of the second section **1302***z* of the transfer ramp **1302** is brought to rest on a top surface of another structure (not shown). Thereafter, cargo can be transferred across a top surface **1302***a* of the transfer ramp **1302**. In some embodiments, at least while transfer ramp **1302** is in the deployed state shown in FIG. **13**E), the handle **1305** can be positioned within a corresponding cutout or recess such that it does not present a roadblock or barrier to transferring cargo across the **1302***a* of the transfer ramp **1302**.

(101) To retract or undeploy the system **1300**, the first and second sections **1302***y* and **1302***z* of the transfer ramp **1302** can be raised and folded against one another such that the transfer ramp **1302** is returned to the position shown in FIG. **13**C. Then, the transfer ramp **1302** can be returned to the stowed position or state beneath the top platform **1322** of the liftgate **1320** (as shown in FIG. **13**A) by sliding the transfer ramp **1302** along the first and second tracks **1306***a* and **1306***b* in a direction generally opposite to the arrow B (FIG. **13**C) and then in a direction generally opposite to the arrow A (FIG. **13**B).

(102) FIG. **14** is a partially schematic cross-sectional side view of another transfer ramp system **1400** ("the system **1400**") configured in accordance with various embodiments of the present technology. As shown, the system **1400** includes a transfer ramp **1402** that can be rolled into a drum **1409** that is stored in a liftgate **1420** or beneath a top platform **1422** of the liftgate **1420**. The transfer ramp **1402** can include a plurality of sections. Three sections **1402***x*, **1402***y*, and **1402***z* of the transfer ramp **1402** are identified individually in FIG. **14**. In some embodiments, the transfer ramp **1402** can include travel limiters (e.g., between immediately adjacent sections of the transfer ramp **1402**) to prevent or hinder the transfer ramp **1402** from drooping when deployed.

(103) To deploy the system **1400**, the transfer ramp **1402** can be pulled to unwind or unroll sections of the transfer ramp **1402** from the drum **1409**. At least some of the sections of the transfer ramp **1402** can then be rested on another structure (not shown) such that the transfer ramp **1402** spans a gap that exists between the top platform **1422** of the liftgate **1420** and the other structure. For example, the section **1402***z* can be laid on a top surface (not shown) of the other structure such that a bottom surface **1402***b* of the section **1402***z* contacts the top surface of the other structure. Thereafter, cargo can be transferred across a top surface **1402***a* of the transfer ramp **1402**. To undeploy or retract the system **1400**, the transfer ramp **1402** can be rewound back into the drum **1409** such that the transfer ramp **1402** is returned to the stowed state or position within the liftgate **1420** or beneath the top platform **1422** of the liftgate **1420**.

(104) FIG. **15**A is a partially schematic cross-sectional side view of another transfer ramp system **1500** ("the system **1500**") configured in accordance with various embodiments of the present technology, and FIGS. **15**B and **15**C are partially schematic side views of slightly alternative implementations of the system **1500**. As shown, the system **1500** includes a transfer ramp **1502** having a top surface **1502***a* and a bottom surface **1502***b*. In a stowed (e.g., retracted, undeployed) state of the system **1500**, the transfer ramp **1502** can be stowed in a corresponding slot or recess **1505** (*i*) within or beneath a floor **1595** (FIGS. **15**A and **15**B) of a bed or box **1590** of a vehicle or trailer and/or (ii) within, beneath, or on a top platform **1522** (FIG. **15**B) of a liftgate **1520**. The transfer ramp **1502** can be manually or mechanically (e.g., via an actuator) moved into and/or out of the recess **1505** generally along or parallel to arrow C. In some embodiments, the system **1500** can include bearings (not shown) that facilitate translating the transfer ramp **1502** generally along or parallel to the arrow C. As best shown in FIGS. **15**B and **15**C, the system **1500** can further include a stop **1571** that prevents the transfer ramp **1502** from being fully removed from the corresponding recess **1505**. When deployed, the transfer ramp **1502** can be moved (e.g., rotated or pivoted) to various positions between position A and position B (FIGS. **15**A and **15**B).

(105) In FIG. **15**A, when deployed, the transfer ramp **1502** can rest on top of the liftgate **1520**. In the alternative implementations shown in FIGS. **15**B and **15**C, the transfer ramp **1502** can be

deployed beneath the top platform **1522** (e.g., a dock plate) of the liftgate **1520**. In some versions of these alternative implementations, the top platform **1522** can rotate or pivot generally along or parallel to arrow D (FIG. **15**B) such that the top platform **1522** can rest on the top surface **1502***a* of the transfer ramp **1502** when the transfer ramp **1502** is deployed and such that the transfer ramp **1502** can pivot to the various positions between position A and position B (FIG. **15**B). In some embodiments, when resting on the top surface **1502***a* of the transfer ramp **1502**, the top platform **1522** can provide a ramped section **1568** (FIG. **15**C) for moving cargo into or out of the vehicle/trailer. Alternatively, the ramped section **1568** shown in FIG. **15**C can be part of the transfer ramp **1502** that is exposed and brought into alignment with the top platform **1522** when the transfer ramp **1502** is deployed. The transfer ramp **1502** can additionally, or alternatively, include a ramped lip portion **1508** in some embodiments to facilitate easier loading and/or unloading of cargo onto and/or off of, respectively, the top surface **1502***a* of the transfer ramp **1502** (e.g., when the bottom surface **1502***b* of the transfer ramp **1502** is contacting a top surface of another structure (not shown)).

(106) Although shown in FIGS. **15**A-**15**C as being fully embedded beneath the floor **1595** of the box **1590** and/or fully beneath the top platform **1522** of the liftgate **1520** when the system **1500** is in a stowed position or state, the top surface **1502***a* of at least a portion of the transfer ramp **1502** can serve as part of the floor **1595** of the box **1590** and/or as the top platform **1522** of the liftgate **1520** in other embodiments of the present technology. For example, referring to FIG. **15**A, a section **1585** of the floor **1595** can be omitted in some embodiments. In these embodiments, the top surface **1502***a* of the transfer ramp **1502** can serve as at least part of the floor for the box **1590**. Continuing with this example, when the transfer ramp **1502** is needed to bridge a gap between (a) the box **1590** or the liftgate **1520** and (b) a top surface of another structure (not shown), the transfer ramp **1502** can be extended outward toward the other structure by moving the transfer ramp **1502** generally along or parallel to the arrow C.

(107) As another example, referring to FIG. **15**A, the floor **1595** of the box **1590** can be divided into two portions: a first section **1585** and a second section **1587**. The first section **1585** of the floor **1595** can be deployed out of the box **1590** to serve as the transfer ramp **1502**. Continuing with this example, in a stowed position or state of the system **1500**, the first section **1585** can sit flush with the second section **1587** of the floor **1595** such that the floor **1595** is generally continuous. When the system **1500** is deployed, the first section **1585** can be moved (e.g., slid or translated) out of the box **1590** until a first portion of the first section **1585** rests on a top surface of another structure (not shown) and/or until a second portion of the first section **1585** rests on the liftgate **1520**. Thereafter, the first section **1585** can be used as a transfer ramp **1502** to move cargo into and/or out of the box **1590**.

(108) FIGS. **16**A-**16**E are partially schematic perspective views of another transfer ramp system **1600** ("the system **1600**") configured in accordance with various embodiments of the present technology. As shown, the system **1600** includes a transfer ramp **1602** that sits on top of a liftgate **1620**. The transfer ramp **1602** includes a top surface **1602***a*, a first side rail **1603***a*, and a second side rail **1603***b*. The first and/or second side rails **1603***a* and **1603***b* can include cutouts **1616***a* and **1616***b* (e.g., notches or slots) through which pegs or protrusions **1621***a* and **1621***b*, respectively, can project when the transfer ramp **1602** is mounted on the liftgate **1620**. The protrusions **1621***a* and **1621***b* can (i) prevent or hinder vertical motion of the transfer ramp **1602** and (ii) limit horizontal motion of the transfer ramp **1602** to the length(s) of the cutout(s) **1616***a* and/or **1616***b*. Although shown with two cutouts **1616***a* and **1616***b* and two corresponding protrusions **1621***a* and **1621***b* in the illustrated embodiment, the system **1600** can include a different number of cutouts **1616** (e.g., one or more than two) and/or a different number of protrusions **1621** (e.g., one or more than two) in other embodiments of the present technology.

(109) Referring now to FIGS. **16**B and **16**C, the system **1600** can be moved from a stowed position or state (as shown in FIG. **16**A) to a deployed position or state (as shown in FIG. **16**C) by moving

or translating the transfer ramp **1602** generally along or parallel to arrow A. In this manner, the transfer ramp **1602** can be extended away from the liftgate **1620** and toward and/or onto a top surface of another structure (not shown). Thereafter, cargo can be transferred across the top surface **1602***a* of the transfer ramp **1602** and/or a top platform **1622** (e.g., a dock plate) of the liftgate **1620**.
(110) In some embodiments, the system **1600** can include a latch or lock (not shown) to hold the transfer ramp **1602** in a stowed position or state (as shown in FIG. **16**A). In these embodiments, the latch can be actuated to release the transfer ramp **1602** before translating the transfer ramp **1602** generally along or parallel to the arrow A. In these and other embodiments, the transfer ramp **1602** can be locked or held in place (e.g., by one or more latches; not shown) at various positions between the stowed position shown in FIG. **16**A and the extended position shown in FIG. **16**C. As such, the transfer ramp **1602** can be used to span gaps of various magnitudes. Alternatively, the transfer ramp **1602** may be locked or held in place only at the stowed position shown in FIG. **16**A and the extended position shown in FIG. **16**C. In these and still other embodiments, the transfer ramp **1602** can be translated a different distance generally along the arrow A at one side of the transfer ramp **1602** than at the other side of the transfer ramp transfer ramp **1602**. This can facilitate the system **1600** accommodating a range of horizontal misalignment between the liftgate **1620** and another structure.
(111) Referring now to FIGS. **16**D and **16**E, the system **1600** can be returned to the stowed position or state (as shown in FIGS. **16**A and **16**E) by moving or translating the transfer ramp **1602** generally along or parallel to arrow B. In embodiments in which the system **1600** includes one or more latches or locks to hold the transfer ramp **1602** in a deployed position, returning the system **1600** to the stowed position or state can include actuating a latch or lock to release the transfer ramp **1602** before moving the transfer ramp **1602** generally along or parallel to the arrow B. As the transfer ramp **1602** returns to the stowed position or state, the transfer ramp **1602** can engage with a latch or lock to retain the transfer ramp **1602** in the stowed position or state.
(112) FIGS. **17**A-**17**C are partially schematic perspective views of another transfer ramp system **1700** ("the system **1700**") configured in accordance with various embodiments of the present technology. The system **1700** includes a transfer ramp **1702** positioned on top of or beneath a top platform **1722** (e.g., a dock plate) of a liftgate **1720**. When positioned beneath the top platform **1722** of the liftgate **1720**, the system **1700** can be mounted to frame rails (not shown) of a vehicle or trailer that includes the liftgate **1720**. Thus, space beneath the liftgate **1720** can be limited and/or a width of the space available for the system **1700** beneath the liftgate **1720** can be less than a width of the top platform **1722** of the liftgate **1720** that is useable for transferring cargo. Therefore, as best shown in FIGS. **17**B and **17**C, the transfer ramp **1702** can include a plurality of sections (identified individually in FIGS. **17**B and **17**C as first section **1702***x*, second section **1702***y*, and third section **1702***z*). The first and third sections **1702***x* and **1702***z* (i) can be folded onto or underneath the second section **1702***y* to reduce the width or footprint of the transfer ramp **1702** for stowage, and (ii) can be unfolded (generally along or parallel to arrows A and B, respectively shown in FIG. **17**B) to expand a width of a top surface **1702***a* (FIGS. **17**B and **17**C) of the transfer ramp **1702** that is usable for transferring cargo across a gap between the top platform **1722** of the liftgate **1720** and another structure (not shown). The first and third sections **1702***x* and **1702***z* can be manually, mechanically, and/or autonomously folded and/or unfolded. Although shown with three sections in the illustrated embodiment, the transfer ramp **1702** can include a different number of sections (e.g., one section, two sections, or more than three sections) in other embodiments of the present technology.
(113) In some embodiments, the transfer ramp **1702** is useable to transfer cargo without deploying the first and third sections **1702***x* and **1702***z*. For example, the transfer ramp **1702** can be deployed as shown in FIG. **17**A, and cargo can be transferred across a bottom surface **1702***b* of the first and third sections **1702***x* and **1702***z*. Widths of other transfer ramps in other embodiments of the present technology, such as widths of the other transfer ramps of the transfer ramp systems illustrated in

FIGS. **1-16**E and/or described herein, can be expandable in a manner generally similar to the width of the transfer ramp **1702** of FIGS. **17**A-**17**C.

(114) FIGS. **18**A-**18**D are partially schematic perspective views of another transfer ramp system **1800** ("the system **1800**") configured in accordance with various embodiments of the present technology. More specifically, FIG. **18**A is a partially schematic perspective view of the system **1800** in a stowed position or state, and FIG. **18**B is a partially schematic perspective view of the system **1800** in an at least partially deployed position or state. The system **1800** is generally similar to the system **100** described above with reference to FIGS. **1-8**L. Similar reference numbers are therefore used in FIGS. **18**A-**24**N as are used in FIGS. **1-8**L to indicate like or at least generally similar components across these drawings, and a detailed discussion of many of these like/similar components of the system **1800** is therefore largely omitted below for the sake of brevity.

(115) As shown, the system **1800** includes a transfer ramp **1802**, a plurality of extendable arms **1804** (identified individually as first extendable arm **1804***a* and second extendable arm **1084***b* in FIGS. **18**A and **18**B), mounting brackets **1806** (identified individually as first through fourth mounting brackets **1806***a*-**1806***d*, respectively), and a ramp latching system **1810** (FIG. **18**B). The system **1800** further includes actuators **1850** (identified individually as first actuator **1850***a*, second actuator **1850***b*, third actuator **1850***c*, and fourth actuator **1850***d* in FIGS. **18**A and **18**B).

(116) In FIG. **18**A, the system **1800** is shown in the stowed position or state beneath a top platform **1822** (e.g., a top plate, a dock plate, a floor of a bed or box) and frame rails **1891** of a vehicle or trailer (e.g., frame rails of the frame of the vehicle/trailer itself, frame rails of a frame of a box positioned on or carried by the vehicle/trailer). More specifically, the transfer ramp **1802** of the system **1800** is shown in the stowed position (a) beneath the top platform **1822** and the frame rails **1891**, (b) above a ramp portion **1894** of a liftgate **1820** of the vehicle/trailer that can be raised or lowered (e.g., to transfer cargo between a box or bed of the vehicle/trailer and the ground in front of the vehicle/trailer), and (c) at a location that does not impede operation of the liftgate **1820**. In some embodiments, the system **1800** can include a bump stop **1892** (*i*) that spaces the liftgate **1820** downwards and increases the space between the top platform **1822** and the ramp portion **1894** of the liftgate **1820** within which the system **1800** can be deployed and retracted, and/or (b) that limits upwards motion of the liftgate **1820** and thereby prevents the liftgate **1820** from colliding with or crushing the system **1800**. For example, as shown in FIG. **18**A, the system **1800** can include a bump stop **1862** on the mounting bracket **1806***c*. The system **1800** may additionally, or alternatively, include a bump stop **1862** on the mounting bracket **1806***d*. The bump stop(s) **1862** can be made of rubber, metal, and/or another suitable material, and/or can be configured to project out from the bracket(s) **1806***c* and/or **1806***d*. Thus, as the liftgate **1820** is raised vertically, a component **1892** on the liftgate **1820** can come into contact with the bump stop(s) **1862** (as opposed to various other components of the system **1800**), and the bump stop(s) **1862** can prevent further vertical motion of the liftgate **1820**.

(117) As best shown in FIGS. **18**A and **18**D, the first extendable arm **1804***a* and/or the second extendable arm **1804***b* can include a curve **1867** (e.g., a bend) or a cutout, such as in an outermost rail bracket of the first extendable arm **1804***a* that is attached to the third mounting bracket **1806***c* and/or in an outermost rail bracket of the second extendable arm **1804***b* that is attached to the fourth mounting bracket **1806***d*. The curve(s) **1867** can facilitate keeping the liftgate **1820** from contacting the first extendable arm **1804***a* and the second extendable arm **1804***b*. Additionally, or alternatively, the liftgate **1820**, when deployed, can be configured to pivot forward slightly (e.g., in a direction generally along the longitudinal axis of the vehicle/trailer). As such, spacers **1828** made of aluminum or another suitable material can be used to offset rubber bumpers **1827** a desired distance from the backmost edge of the top platform **1822** to, for example, provide a clearance for deployment of the liftgate **1820** and/or the transfer ramp **1802** of the system **1800**. For example, the spacers **1828** can be used to offset the rubber bumpers **1827** by approximately 25.4 cm (10 inches) or less, such as by approximately 12.7 cm (5 inches), approximately 7.62 cm (3 inches),

approximately 5.08 cm (2 inches), approximately 2.54 cm (1 inch), approximately 1.27 cm (0.5 inch), or another suitable distance. Therefore, when the rubber bumpers **1827** are backed against another structure (e.g., another vehicle; a trailer; a loading dock; or rubber bumpers of another vehicle, trailer, or loading dock), a gap can exist between the back edge of the top platform **1822** and a top surface of the other structure onto which cargo can be placed. Continuing with this example, the gap can be approximately 50 cm (e.g., 52 cm, 50.8 cm (20 inches), 48 cm) or less in width, such as approximately 38.1 cm (15 inches), approximately 33.02 cm (13 inches), approximately 30.48 cm (12 inches), approximately 27.94 cm (11 inches), approximately 26.67 cm (10.5 inches), approximately 25.4 cm (10 inches), or less in width.

(118) The mounting brackets **1806** of the system **1800** are fixedly attached (i) to the top platform **1822** and/or (ii) to the vehicle/trailer such that the system **1800** is mounted to or under the frame rails **1891** of the vehicle/trailer. Referring to FIGS. **18**C and **18**D, the third mounting bracket **1806***c* can be (a) positioned within a space proximate to a first side of the liftgate **1820** and/or (b) fixedly attached directly or indirectly to the top platform **1822**, such as via sides (not shown) similar to the sides **126** of the system **100** of FIGS. **1-8**L. The fourth mounting bracket **1806***d* (FIG. **18**A) can be similarly (a) positioned within a space proximate to a second side of the liftgate **1820** opposite the first side and/or (b) fixedly attached directly or indirectly to the top platform **1822**.

(119) FIGS. **19**A and **19**B are additional partially schematic perspective views of the system **1800**. Referring to FIGS. **18**A, **18**B, **19**A, and **19**B together, the first mounting bracket **1806***a* can be positioned beneath the frame rails **1891** of the vehicle/trailer and at a location proximate to a first side of the vehicle/trailer. In some embodiments, the first mounting bracket **1806***a* can be fixedly attached to the vehicle/trailer to or under the frame rails **1891** of the vehicle/trailer, such as using one or more upper clamps **1816**. The second mounting bracket **1806***b* (FIGS. **18**A and **18**B) can similarly be positioned at a location proximate to a second side of the vehicle/trailer opposite the first side, and/or can be fixedly attached to or under the frame rails **1891** of the vehicle/trailer, such as using one or more upper clamps **1816**. As discussed above, the frame rails **1891** can include frame rails of the frame of the vehicle/trailer itself and/or frame rails of a frame of a box positioned on or carried by (e.g., the frame or chassis of) the vehicle/trailer. As a specific example, the first and second mounting brackets **1806***a* and **1806***b* of the system **1800** can be fixedly attached to the vehicle/trailer such that the system **1800** is mounted to, on, within, and/or beneath a chassis of the vehicle/trailer. Mounting of the system **1800** to the outer portions of the vehicle/trailer can facilitate using a wider transfer ramp **1802** (e.g., a transfer ramp **1802** that spans an entirety, a substantial majority, and/or nearly all of the width of the box or bed of the vehicle/trailer) than the transfer ramp **102** used in the system **100** of FIGS. **1-8**L. Although shown as separate mounting brackets **1806***a***-1806***d* in the illustrated embodiment, the first mounting bracket **1806***a* can be integrated with the third mounting bracket **1806***c* in other embodiments of the present technology, and/or the second mounting bracket **1806***b* can be integrated with the fourth mounting bracket **1806***d* in other embodiments of the present technology. Additionally, or alternatively, the first mounting bracket **1806***a* can be integrated with the second mounting bracket **1806***b*, and/or the third mounting bracket **1806***c* can be integrated with the fourth mounting bracket **1086***d*.

(120) With continuing reference to FIGS. **18**A, **18**B, **19**A, and **19**B together, the first actuator **1850***a* of the system **1800** can be attached at one end to the first mounting bracket **1806***a* and can be attached at an opposite end to a distal portion of the first extendable arm **1804***a*. In some embodiments, the first actuator **1850***a* can be a pneumatic, hydraulic, electric (e.g., electro-hydraulic, electromechanical), and/or piezoelectric linear actuator. As discussed in greater detail below, the first actuator **1850***a* can be configured to extend and retract generally along its longitudinal axis (e.g., generally along or parallel to the arrow A illustrated in FIG. **18**B) and thereby extend and retract, respectively, the first extendable arm **1804***a* (e.g., generally along a longitudinal axis of the first extendable arm **1804***a*). Similarly, the second actuator **1850***b* (not shown in FIGS. **19**A and **19**B) of the system **1800** can be attached at one end to the second

mounting bracket **1806***b* and can be attached at an opposite end to a distal portion of the second extendable arm **1804***b*. In some embodiments, the second actuator **1850***b* can be a pneumatic, hydraulic, electric (e.g., electro-hydraulic, electromechanical), and/or piezoelectric linear actuator. As discussed in greater detail below, the second actuator **1850***b* can be configured to extend and retract generally along its longitudinal axis (e.g., generally along or parallel to the arrow A illustrated in FIG. **18**B) and thereby extend and retract, respectively, the second extendable arm **1804***b* (e.g., generally along a longitudinal axis of the second extendable arm **1804***b*). In some embodiments, the first actuator **1850***a* and the second actuator **1850***b* can be extended or retracted in unison and/or independently of one another. In other embodiments of the present technology, the system **1800** can omit the first actuator **1850***a* or the second actuator **1850***b* such that the system **1800** uses a single actuator **1850** to extend and retract the first and second extendable arms **1804***a* and **1804***b* along their respective longitudinal axis.

(121) The third actuator **1850***c* of the system **1800** can be attached at one end to the first mounting bracket **1806***a* and can be attached at an opposite end to a proximal portion of the first extendable arm **1804***a*. In some embodiments, the third actuator **1850***c* can be a pneumatic, hydraulic, electric (e.g., electro-hydraulic, electromechanical), and/or piezoelectric linear actuator. As discussed in greater detail below, the third actuator **1850***c* can be configured to extend and retract generally along its longitudinal axis (e.g., generally along or parallel to the arrow B illustrated in FIG. **18**B) and thereby move (e.g., translate, pivotably move, rotate, rotatably move, adjust a pitch of) the first extendable arm **1804***a*, such as by vertically lowering and/or raising the proximal portion of first extendable arm **1804***a*. Similarly, the fourth actuator **1850***d* (not shown in FIGS. **19**A and **19**B) of the system **1800** can be attached at one end to the second mounting bracket **1806***b* and can be attached at an opposite end to a proximal portion of the second extendable arm **1804***b*. In some embodiments, the fourth actuator **1850***d* can be a pneumatic, hydraulic, electric (e.g., electro-hydraulic, electromechanical), and/or piezoelectric linear actuator. As discussed in greater detail below, the fourth actuator **1850***d* can be configured to extend and retract generally along its longitudinal axis (e.g., generally along or parallel to the arrow B illustrated in FIG. **18**B) and thereby move (e.g., translate, pivotably move, rotate, rotatably move, adjust a pitch of) the second extendable arm **1804***b*, such as by vertically lowering and/or raising the proximal portion of second extendable arm **1804***b*. In some embodiments, the third actuator **1850***c* and the fourth actuator **1850***d* can be extended or retracted in unison and/or independently of one another. In these and other embodiments, the third actuator **1850***c* and/or the fourth actuator **1850***d* can be extended or retracted (a) in unison with the first actuator **1850***a* and/or the second actuator **1850***b*, and/or (b) independently of the first actuator **1850***a* and/or the second actuator **1850***b*. In some embodiments of the present technology, the system **1800** can omit the third actuator **1850***c* or the fourth actuator **1850***d* such that the system **1800** uses a single actuator **1850** to vertically raise and lower the proximal portions of the first and second extendable arms **1804***a* and **1804***b*.

(122) FIG. **20** is a perspective view of the transfer ramp system **1800** of FIGS. **18**A-**19**B in a deployed state (e.g., a fully deployed state). As shown, the transfer ramp **1802** has been (a) maneuvered through the space beneath the top platform **1822** and above the liftgate **1820**, and through a gap between the top platform **1822** and a top surface **232** (e.g., a top platform, top plate, dock plate) of another structure **230** (e.g., another vehicle/trailer), and (b) lowered until the transfer ramp **1802** rests on the top platform **1822** and the top surface **232**. In this state, the transfer ramp **1802** of the system **1800** can be used to transfer cargo between the vehicle/trailer including the system **1800** and the vehicle/trailer including the top surface **232**.

(123) FIG. **21** is a partially schematic side view illustrating motion of the transfer ramp system **1800** of FIGS. **18**A-**20**. As shown in FIG. **21** and as described in greater detail below, the transfer ramp **1802** can be moved generally along the arrow A shown in FIG. **21** (e.g., along a generally ‘S’-shaped trajectory) to deploy and retract the transfer ramp **1802** through the gap that exists between the top platform **1822** and the top surface **232**. More specifically, user controls (e.g.,

mounted to and/or stowed on the liftgate **1820** and/or the vehicle/trailer) and/or a microcontroller can be used to control or coordinate operation of the actuators **1850** to move the transfer ramp **1802** generally along the arrow A. In some embodiments, the arrow A can represent a predetermined path. For example, the arrow A can represent a path that reduces or minimizes a gap that must exist between a backmost edge of the top plate **1822** and a backmost edge of the top surface **232** to deploy and/or retract the transfer ramp **1802** of the system **1800** (e.g., without contacting the top platform **1822** and/or the top surface **232**). For example, the arrow A can represent an optimized or tightest deployment/retraction path for the transfer ramp **1802**. As discussed in greater detail below, while deploying or retracting the transfer ramp **1802**, the system **1800** can be configured to (a) monitor potentiometer readings associated with the actuators **1850***a*-**1850***d* and (b) control one or more of the actuators **1850***a*-**1850***d* such that the transfer ramp **1802** is deployed or retracted generally along the predetermined path represented by the arrow A.

(124) FIGS. **22**A-**22**N are partially schematic side views of the transfer ramp system **1800** of FIGS. **18**A-**21** illustrating a method of operating the transfer ramp system **1800** in accordance with various embodiments of the present technology. In addition, FIGS. **23**A-**23**N are other partially schematic side views of the transfer ramp system of FIGS. **18**A-**18**D illustrating the method of operating the transfer ramp system **1800** shown in FIGS. **22**A-**22**N, and FIGS. **24**A-**24**N are partially schematic, cross-sectional side views of the transfer ramp system **1800** of FIGS. **18**A-**18**D (without the transfer ramp **1802** illustrated) illustrating the method of operating the transfer ramp system shown in FIGS. **22**A-**22**N and in FIGS. **23**A-**23**N. All or a subset of the steps of the method illustrated in FIGS. **22**A-**24**N can be performed in accordance with the discussion of the system **100** and/or the system **1800** above.

(125) Referring to FIGS. **22**A, **23**A, and **24**A together, the method begins with the system **1800** in the stowed position beneath the top platform **1822** and above the liftgate **1820**. More specifically, the vehicle/trailer can initially be backed toward the another structure **230** until (a) the vehicle/trailer is positioned such that rubber bumpers **1827** face and/or contact the other structure **230**, and/or (b) the top platform **1822** is spaced a distance from the top surface **232** of the other structure **230** that is less than a length of the transfer ramp **1802** measured in a direction generally along a longitudinal axis of the vehicle/trailer. As discussed above, the other structure **230** can be a commercial vehicle, a loading dock, a trailer, a bed of a truck, or another structure. When the vehicle/trailer that includes the system **1800** is positioned as shown in FIGS. **22**A, **23**A, and **24**A, a gap G (FIG. **22**A) exists between an edge (e.g., a backmost edge) of the top platform **1822** and an edge of the top surface **232** of the other structure **230**.

(126) Referring now to FIGS. **22**B, **23**B, and **24**B together, the method continues by extending the first and second extendable arms **1804***a* and **1804***b* and thereby extending or moving the transfer ramp **1802** out from a space under the top platform **1822** and above the liftgate **1820**. More specifically, as best shown in FIGS. **23**B and **24**B, the first actuator **1850***a* and/or the second actuator **1850***b* can be extended in a direction generally along or parallel to arrow A. Extension of the first actuator **1850***a* and/or the second actuator **1850***b* in the direction generally along or parallel to the arrow A can extend the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow B (e.g., by exerting a pushing force on the distal portions of the first and second extendable arms **1804***a* and **1804***b*). During this process, the third actuators **1850***c* and/or the fourth actuator **1850***d* may be stationary or may be actuated (e.g., extended and/or retracted, such as at a same or different rate from the first and second actuators **1850***a* and **1850***b*) in addition to and/or while the first and second actuators **1850***a* and/or **1850***d*, respectively, are extended.

(127) Referring now to FIGS. **22**C, **23**C, and **24**C together, the method continues by rotating the first and second extendable arms **1804***a* and **1804***b*. More specifically, as best shown in FIGS. **23**C and **24**C, the third actuator **1850***c* and/or the fourth actuator **1850***d* can be extended in a direction generally along or parallel to arrow C. Extension of the third actuator **1850***c* and/or the fourth

actuator **1850***d* in the direction generally along or parallel to the arrow C can, at least while the first actuator **1850***a* and/or the second actuator **1850***b*, respectively, are stationary and/or extended at a slower rate, (i) rotate and/or vertically lower proximal portions of the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow D, (ii) adjust (e.g., increase) a pitch of the first and second extendable arms **1804***a* and **1804***b* and/or of the transfer ramp **1802**, and/or (iii) pivot the first and second extendable arms **1804***a* and **1804***b* (e.g., generally about a point at which the first and second extendable arms **1804***a* and **1804***b* are operably coupled to the first and second actuators **1850***a* and **1850***b*). In some embodiments, the first actuator **1850***a* and/or the second actuator **1850***b* may optionally be retracted (e.g., at a same or different rate than the third and fourth actuators **1850***c* and **1850***d*) in addition to and/or while the third and fourth actuators **1850***c* and **1850***d*, respectively, are extended. As (i) the third and fourth actuators **1850***c* and **1850***d* are extended while (ii) the first and second actuators **1850***a* and **1850***b* are stationary, extended at a slower rate, or are retracted, (a) the distal portions of the first and second extendable arms **1804***a* and **1804***b* and/or (b) a back lip/edge of the transfer ramp **1802** can be moved (e.g., rotated, tilted, pivoted, vertically raised) in a direction generally along or parallel to arrow E.

(128) Rotating (e.g., pivoting, tilting) the first and second extendable arms **1804***a* and **1804***b* and/or adjusting the pitch of the first and second extendable arms **1804***a* and **1804***b* in this manner can facilitate reducing, narrowing, or minimizing a distance required between the transfer ramp **1802** and the backmost edge of the top platform **1822** while deploying the transfer ramp **1802**. In other words, rotating and/or adjusting the pitch of the first and second extendable arms **1804***a* and **1804***b* in this manner can facilitate (a) deploying the transfer ramp **1802** along an optimized or tightest possible path relative to the top platform **1822** and/or (b) minimizing, narrowing, or reducing a minimum possible gap G (FIG. **22**A) that is required between the top platform **1822** and the top surface **232** in order to deploy the transfer ramp **1802** (e.g., without contacting the top platform **1822** and/or the top surface **232**). As a specific example, rotating and/or adjusting the pitch of the first and second extendable arms **1804***a* and **1804***b* can obviate a need for the structure **230** to include bumpers and/or spacers (e.g., similar to the bumpers **1827** and/or spacers **1828** installed on the vehicle/trailer that includes the system **1800**).

(129) Referring now to FIGS. **22**D, **23**D, and **24**D together, the method continues by further extending the first and second extendable arms **1804***a* and **1804***b* and thereby further extending or moving the transfer ramp **1802** outward. More specifically, as best shown in FIGS. **23**D and **24**D, the first actuator **1850***a* and/or the second actuator **1850***b* can be extended in a direction generally along or parallel to arrow F. Extension of the first actuator **1850***a* and/or the second actuator **1850***b* in the direction generally along or parallel to the arrow F can extend the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow G (e.g., by exerting a pushing force on the distal portions of the first and second extendable arms **1804***a* and **1804***b*). In this manner, the transfer ramp **1802** can be moved up through the gap G (FIG. **22**A) between the top platform **1822** and the top surface **232**. During this process, the third actuators **1850***c* and/or the fourth actuator **1850***d* may be stationary or may be actuated (e.g., extended and/or retracted, such as at a same or different rate from the first and second actuators **1850***a* and **1850***b*) in addition to and/or while the first and second actuators **1850***a* and/or **1850***d*, respectively, are extended.

(130) Referring now to FIGS. **22**E, **23**E, and **24**E together, the method continues by rotating or pivoting the transfer ramp **1802** from the stowed orientation/stored arrangement (FIGS. **22**D and **23**D) to a deployed orientation/deployed arrangement (e.g., in which the transfer ramp **1802** is relatively flat or horizontal, and/or in which top surface **1802***a* (FIG. **23**G) and/or a bottom surface **1802***b* (FIGS. **22**E and **23**E) of the transfer ramp **1802** is/are brought more into alignment with the top platform **1822** and/or the top surface **232** of the other structure **230**). In some embodiments, rotating or pivoting the transfer ramp **1802** can include rotating or pivoting the transfer ramp **1802** in a direction generally along or parallel to arrow H. In some embodiments, rotating or pivoting the transfer ramp **102** can include releasing a latchable portion (e.g., a latchable portion **2509**

illustrated in FIGS. **25**A-**25**D, a latchable portion **2609** illustrated in **26**A-**26**E, and/or a latchable portion **2909** of FIGS. **29**A and **29**B) of the transfer ramp **1802** from a release latch (e.g., a cable release latch **2511** illustrated in FIGS. **25**A-**25**D, a cable release latch **2611** illustrated in FIGS. **26**A-**26**E, and/or a first notch **2914***a* illustrated in FIGS. **29**A and **29**B) of the ramp latching system **1810**, such as by actuating the release latch via an actuation mechanism (e.g., a thumb latch shown in FIG. **22**E) and/or a cable (not shown). In these and other embodiments, rotating or pivoting the transfer ramp **1802** can include rotating or pivoting the transfer ramp **1802** using one or more springs (e.g., one or more torsion springs) and/or by extending struts **1807** (FIGS. **26**A-**26**E) (e.g., such that the transfer ramp **1802** pivots about hinge pins and/or points at which the first and second extendable arms **1804***a* and **1804***b* are attached to the transfer ramp **1802**). In these and other embodiments, rotating or pivoting the transfer ramp **1802** can include pushing on the transfer ramp **1802** or otherwise manually rotating or pivoting the transfer ramp **1802**.

(131) Referring next to FIGS. **22**F, **23**F, and **24**F together, the method continues by retracting the first and second extendable arms **1804***a* and **1804***b* and thereby retracting or lowering the transfer ramp **1802** toward the top platform **1822** and the top surface **232** while the transfer ramp **1802** is in the generally flat or horizonal orientation (or in another orientation that is more in alignment with the top platform **1822** of the liftgate **120** than the stowed orientation of the transfer ramp **1802**). More specifically, as best shown in FIGS. **23**F and **24**F, the first actuator **1850***a* and/or the second actuator **1850***b* can be retracted in a direction generally along or parallel to arrow I. Retraction of the first actuator **1850***a* and/or the second actuator **1850***b* in the direction generally along or parallel to the arrow I can retract the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow J (e.g., by exerting a pulling force on the distal portions of the first and second extendable arms **1804***a* and **1804***b*). During this process, the third actuators **1850***c* and/or the fourth actuator **1850***d* may be stationary or may be actuated (e.g., extended and/or retracted, such as at a same or different rate from the first and second actuators **1850***a* and **1850***b*) in addition to and/or while the first and second actuators **1850***a* and/or **1850***d*, respectively, are retracted. As (i) the first and second actuators **1850***a* and **1850***b* are retracted while (ii) the third and fourth actuators **1850***c* and **1850***d* are stationary and/or actuated, the transfer ramp **1802** can be lowered such that at least a portion of the bottom surface **1802***b* of the transfer ramp **1802** is positioned above the top surface **232** of the other structure **230** and at least another portion of the bottom surface **1802***b* is positioned above the top platform **1822**. In these and other embodiments, lowering the transfer ramp **1802** can include lowering the transfer ramp **1802** until a portion of the bottom surface **1802***b* of the transfer ramp **1802** rests on the top surface **232** of the other structure **230**, another portion of the bottom surface **1802***b* rests on the top platform **1822**, and/or the transfer ramp **1802** spans (e.g., bridges) the gap G (FIG. **22**A).

(132) Referring now to FIGS. **22**G, **23**G, and **24**G together, the method optionally continues by rotating or pivoting the first and second extendable arms **1804***a* and **1804***b*. More specifically, as best shown in FIGS. **23**G and **24**G, the third actuator **1850***c* and/or the fourth actuator **1850***d* can be retracted in a direction generally along or parallel to arrow K. Retraction of the third actuator **1850***c* and/or the fourth actuator **1850***d* in the direction generally along or parallel to the arrow K can, at least while the first actuator **1850***a* and/or the second actuator **1850***b* are stationary or retracting at a slower rate, (i) rotate and/or vertically raise proximal portions of the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow L, (ii) adjust (e.g., decrease) a pitch of the first and second extendable arms **1804***a* and **1804***b*, and/or (iii) pivot the first and second extendable arms **1804***a* and **1804***b* (e.g., generally about the point at which the first and second extendable arms **1804***a* and **1804***b* are operably coupled to the first and second actuators **1850***a* and **1850***b*). In some embodiments, the first actuator **1850***a* and/or the second actuator **1850***b* may optionally be extended in addition to and/or while the third and fourth actuators **1850***c* and **1850***d*, respectively, are retracted. As the third and fourth actuators **1850***c* and **1850***d* are retracted while the first and second actuators **1850***a* and **1850***b* are stationary, retracted at

a slower rate, or are extended, a force can be exerted on distal portions of the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow M, and/or the distal portion of the first and/or second extendable arms **1804***a* and **1804***b* can be rotated, pivoted, and/or vertically lowered in the direction generally along or parallel to the arrow M.
(133) Rotating and/or retracting the first and second extendable arms **1804***a* and **1804***b* in this manner can exert forces on the transfer ramp **1802** and/or on the corresponding hinges/pivot points connecting the transfer ramp **1802** to the first and second extendable arms **1804***a* and **1804***b*. In turn, these forces can bend (e.g., deform, warp) the transfer ramp **1802** such that the transfer ramp **1802** is conformed to the top platform **1822** and/or the top surface **232** (e.g., to deal with any misalignment and/or to provide a suitably flat surface across which cargo can be transferred, as discussed above with reference to the system **100** of FIGS. **1-8**L). For example, as portions of the transfer ramp **1802** contact the top platform **1822** and/or the top surface **232**, corresponding actuators **1850** can overcurrent and cease extending or retracting while other ones of the actuators **1850** continue to extend or retract, thereby conforming the transfer ramp **1802** to the arrangement of the top platform **1822** and/or the top surface **232**.
(134) Additionally, or alternatively, the forces exerted on the transfer ramp **1802** and/or on the corresponding hinges/pivot points from tilting and/or retracting the first and second extendable arms **1804***a* and **1804***b* can be used to overcome misalignment (e.g., between opposite sides of the top platform **1822**, between opposite sides of the top surface **232**, and/or between the top platform **1822** and the top surface **232**) and/or to change the height of (i) top platform **1822** on the vehicle/trailer that includes the system **1800** and/or (ii) the top surface **232** of the other structure **230**. For example, tilting and/or retracting the first and second extendable arms **1804***a* and **1804***b* after the transfer ramp **1802** initially contacts the top platform **1822** and the top surface **232** can press or smash the transfer ramp **1802** into the top platform **1822** and the top surface **232** to bring the top platform **1822** and the top surface **232** into closer alignment with one another and/or to bring the transfer ramp **102** closer to a relatively flat, horizonal, or level orientation. As a specific example, consider a scenario in which the top platform **1822** is initially positioned at a lower height than the top surface **232**. In this scenario, titling and/or retracting the first and second extendable arms **1804***a* and **1804***b* beyond the point at which the transfer ramp **1802** initially contacts the top platform **1822** and the top surface **232** can lower the height of the top surface **232** (e.g., via the suspension of a vehicle corresponding to the other structure **230** that includes the top surface **232**) and/or can raise the height of the top platform **1822** (e.g., via an upwards force that results as the transfer ramp **1802** is pushed down into the top surface **232**). As a result, the height of the top platform **1822** can be brought into alignment with the height of the top surface **232**, meaning that the transfer ramp **1802** can be brought to (or at least closer to) a generally flat, horizontal, or level orientation thereby making transfers of cargo across the transfer ramp **1802** easier.
(135) As another specific example, consider a scenario in which a first side of the top platform **1822** (e.g., proximate the first extendable arm **1804***a*) is positioned higher than a second side of the top platform **1822** (e.g., proximate the second extendable arm **1804***b*). When a first side of the transfer ramp **1802** makes initial contact with the first side of the top platform **1822** and the second side of the transfer ramp **1802** is not in contact with the second side of the top platform **1822**, both the first and second extendable arms **1804***a* and **1804***b* can continue to be retracted. In some embodiments, this can force the first side of the transfer ramp **1802** down into the first side of the top platform **1822** to lower the height of the first side of the top platform **1822** relative to the second side of the top platform **1822** (e.g., using the suspension of the vehicle/trailer that includes the system **1800**). Lowering the height of the first side of the top platform **1822** relative to the second side of the top platform **1822** can bring the second side of the transfer ramp **1802** into contact with the second side of the top platform **1822**, thereby (a) leveling the transfer ramp **1802** and/or the top platform **1822** and/or (b) eliminating, reducing, or minimizing a roadblock or obstacle posed by the second side of the transfer ramp **1802** to transferring cargo across at least the

second side of the transfer ramp **1802**.

(136) With the transfer ramp **1802** positioned as shown in FIG. **22**G, the method continues by transferring cargo across the top surface **1802***a* (FIG. **23**G) of the transfer ramp **102**. For example, cargo (e.g., a pallet or other object) can be loaded into the vehicle/trailer by transferring the cargo from the top surface **232** of the other structure **230**, across the top surface **1802***a* of the transfer ramp **1802**, and onto the top platform **1822**. Additionally, or alternatively, cargo can be unloaded from the vehicle/trailer by transferring the cargo from the top platform **1822**, across the top surface **1802***a* of the transfer ramp **1802**, and onto the top surface **232** of the other structure **230**.

(137) Referring next to FIGS. **22**H, **23**H, and **24**H together, the method optionally continues by tilting the first and second extendable arms **1804***a* and **1804***b*. More specifically, as best shown in FIGS. **23**H and **24**H, the third actuator **1850***c* and/or the fourth actuator **1850***d* can be extended in a direction generally along or parallel to arrow N. Extension of the third actuator **1850***c* and/or the fourth actuator **1850***d* in the direction generally along or parallel to the arrow N can, at least while the first actuator **1850***a* and/or the second actuator **1850***b*, respectively, are stationary and/or extended at a slower rate, (i) rotate and/or vertically lower proximal portions of the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow O, (ii) adjust (e.g., increase) a pitch of the first and second extendable arms **1804***a*, and/or (iii) pivot the first and second extendable arms **1804***a* and **1804***b* (e.g., generally about a point at which the first and second extendable arms **1804***a* and **1804***b* are operably coupled to the first and second actuators **1850***a* and **1850***b*). In some embodiments, the first actuator **1850***a* and/or the second actuator **1850***b* may optionally be retracted (e.g., at a same or different rate than the third and fourth actuators **1850***c* and **1850***d*) in addition to and/or while the third and fourth actuators **1850***c* and **1850***d*, respectively, are extended. As (i) the third and fourth actuators **1850***c* and **1850***d* are extended while (ii) the first and second actuators **1850***a* and **1850***b* are stationary, extended (e.g., at a same or slower rate than the third and fourth actuators **1850***c* and **1850***d*), or are retracted, the forces exerted (e.g., on the transfer ramp **1802** and/or on the corresponding hinges/pivot points connecting the transfer ramp **1802** to the first and second extendable arms **1804***a* and **1804***b*) at the step of the method illustrated in FIGS. **22**G, **23**G, and **24**G above can be lessened or removed, and/or the distal portion of the first and/or second extendable arms **1804***a* and **1804***b* can be moved (e.g., rotated, tilted, pivoted, vertically raised) generally along the arrow P. In embodiments in which the transfer ramp **1802** is resiliently deformable, removal of the forces can permit the transfer ramp **1802** to return to or toward its initial or default state prior to application of the forces at the step illustrated by FIGS. **22**G, **23**G, and **24**G.

(138) Referring now to FIGS. **22**I, **23**I, and **24**I together, the method continues by extending the first and second extendable arms **1804***a* and **1804***b* and thereby extending or moving the transfer ramp **1802** outward. More specifically, as best shown in FIGS. **23**I and **24**I, the first actuator **1850***a* and/or the second actuator **1850***b* can be extended in a direction generally along or parallel to arrow Q. Extension of the first actuator **1850***a* and/or the second actuator **1850***b* in the direction generally along or parallel to the arrow Q can extend the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow R (e.g., by exerting a pushing force on the distal portions of the first and second extendable arms **1804***a* and **1804***b*). In this manner, the transfer ramp **1802** can be raised above the top platform **1822** and the top surface **232**. During this process, the third actuators **1850***c* and/or the fourth actuator **1850***d* may be stationary or may be actuated (e.g., extended and/or retracted, such as at a same or different rate from the first and second actuators **1850***a* and **1850***b*) in addition to and/or while the first and second actuators **1850***a* and/or **1850***d*, respectively, are extended.

(139) Referring now to FIGS. **22**J, **23**J, and **24**J together, the method continues by rotating or pivoting the transfer ramp **1802** to the stowed orientation. In some embodiments, rotating or pivoting the transfer ramp **1802** to the stowed orientation can include rotating or pivoting the transfer ramp **1802** in a direction generally along or parallel to the arrow S. In these and other

embodiments, rotating or pivoting the transfer ramp **1802** can include rotating or pivoting the transfer ramp **1802** using a transfer ramp storage system, such as a transfer ramp storage system discussed in greater detail below with reference to FIGS. **25**A-**25**D and/or a transfer ramp storage system discussed in greater detail below with reference to FIGS. **26**A-**26**E. In these and still other embodiments, rotating or pivoting the transfer ramp **1802** can include rotating or pivoting the transfer ramp **1802** using one or more springs (e.g., one or more torsion springs), and electric motor, and/or by retracting struts **1807** (FIGS. **26**A-**26**E) (e.g., such that the transfer ramp **1802** pivots about hinge pins and/or points at which the first and second extendable arms **1804***a* and **1804***b* are attached to the transfer ramp **1802**). In these and other embodiments, rotating or pivoting the transfer ramp **1802** can include pushing on the transfer ramp **1802** or otherwise manually rotating or pivoting the transfer ramp **1802** toward the stowed orientation and/or until a latchable portion of the transfer ramp **1802** is engaged with and/or held by a latch.

(140) Referring now to FIGS. **22**K, **23**K, and **24**K together, the method continues by retracting the first and second extendable arms **1804***a* and **1804***b* and thereby retracting or lowering the transfer ramp **1802** in the stowed orientation and/or down through the gap G between the top platform **1822** and the top surface **232** of the other structure **230**. More specifically, as best shown in FIGS. **23**K and **24**K, the first actuator **1850***a* and/or the second actuator **1850***b* can be retracted in a direction generally along or parallel to arrow T. Retraction of the first actuator **1850***a* and/or the second actuator **1850***b* generally along or parallel to the arrow T can retract the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow U (e.g., by exerting a pulling force on the distal portions of the first and second extendable arms **1804***a* and **1804***b*). During this process, the third actuators **1850***c* and/or the fourth actuator **1850***d* may be stationary or may be actuated (e.g., extended and/or retracted, such as at a same or different rate from the first and second actuators **1850***a* and **1850***b*) in addition to and/or while the first and second actuators **1850***a* and/or **1850***d*, respectively, are retracted.

(141) Referring now to FIGS. **22**L, **23**L, and **24**L together, the method continues by rotating the first and second extendable arms **1804***a* and **1804***b*. More specifically, as best shown in FIGS. **23**L and **24**L, the third actuator **1850***c* and/or the fourth actuator **1850***d* can be retracted in a direction generally along or parallel to arrow V. Retraction of the third actuator **1850***c* and/or the fourth actuator **1850***d* in the direction generally along or parallel to the arrow V can, at least while the first actuator **1850***a* and/or the second actuator **1850***b*, respectively, are stationary and/or retracted at a slower rate, (i) rotate and/or vertically raise proximal portions of the first and second extendable arms **1804***a* and **1804***b* in a direction generally along or parallel to arrow W, (ii) adjust (e.g., decrease) a pitch of the first and second extendable arms **1804***a* and **1804***b* and/or of the transfer ramp **1802**, and/or (iii) pivot the first and second extendable arms **1804***a* and **1804***b* (e.g., generally about a point at which the first and second extendable arms **1804***a* and **1804***b* are operably coupled to the first and second actuators **1850***a* and **1850***b*). In some embodiments, the first actuator **1850***a* and/or the second actuator **1850***b* may optionally be extended (e.g., at a same or different rate than the third and fourth actuators **1850***c* and **1850***d*) in addition to and/or while the third and fourth actuators **1850***c* and **1850***d*, respectively, are extended. As (i) the third and fourth actuators **1850***c* and **1850***d* are retracted while (ii) the first and second actuators **1850***a* and **1850***b* are stationary, retracted at a slower rate, or are extended, (a) the distal portions of the first and second extendable arms **1804***a* and **1804***b* and/or (b) the back lip/edge of the transfer ramp **1802** can be moved (e.g., rotated, tilted, pivoted, vertically lowered) in a direction generally along or parallel to arrow X.

(142) Referring next to FIGS. **22**M, **23**M, and **24**M together, the method continues by further retracting the first and second extendable arms **1804***a* and **1804***b* and thereby further retracting or lowering the transfer ramp **1802**. More specifically, as best shown in FIGS. **23**M and **24**M, the first actuator **1850***a* and/or the second actuator **1850***b* can be retracted in a direction generally along or parallel to arrow Y. Retraction of the first actuator **1850***a* and/or the second actuator **1850***b* in the direction generally along or parallel to the arrow Y can retract the first and second extendable arms

1804*a* and **1804***b* in a direction generally along or parallel to arrow Z (e.g., by exerting a pulling force on the distal portions of the first and second extendable arms **1804***a* and **1804***b*). During this process, the third actuators **1850***c* and/or the fourth actuator **1850***d* may be stationary or may be actuated (e.g., extended and/or retracted, such as at a same or different rate from the first and second actuators **1850***a* and **1850***b*) in addition to and/or while the first and second actuators **1850***a* and/or **1850***d*, respectively, are retracted.

(143) Referring now to FIGS. **22**N, **23**N, and **24**N together, the method continues by returning the system **1800** to the stowed position or state beneath the top platform **1822** and above the liftgate **1820**. In some embodiments, the system **1800** can include bump stops and/or storage/stowage stops for the transfer ramp **1802**. For example, returning the system **1800** to the stowed position can include retracting the first and second extendable arms **1804***a* and **1804***b* (and/or controlling the third actuator **1850***c* and/or the fourth actuator **1850***d*) until the transfer ramp **1802** contacts bump stops that are composed of rubber and/or another suitable material, and/or that are positioned beneath the top platform **1822** of the liftgate **1820**. In this manner, the bump stops can limit further inward motion of the transfer ramp **1802** (e.g., in a direction generally toward the front of the vehicle/trailer) beneath the top platform **1822**. As another example, the system **1800** can include a storage/stowage stop that is composed of rubber or another suitable material and/or that may function at least partially as a bump stop.

(144) FIGS. **34**A-**34**C are partially schematic perspective views of the system **1800** illustrating an example of such a storage/stowage stop. Components of this stowage stop are also shown in FIG. **22**N. Referring to FIGS. **34**A and **34**B, the stowage stop includes three stowage brackets **3436** (identified individually in FIGS. **34**A and **34**B as first stowage bracket **3436***a*, second stowage bracket **3436***b*, and third stowage bracket **3436***c*). As shown, the stowage brackets **3436***a*-**3436***c* are positioned (i) at least partially above the liftgate **1820** and (ii) beneath the top platform **1822** and/or the frame rails **1891** (FIG. **22**N) of the vehicle/trailer. In some embodiments, the stowage brackets **3436** can be mounted directly or indirectly to/beneath the top platform **1822** and/or the frame rails **1891** of the vehicle/trailer, such as using upper clamps **3437** (FIG. **22**N) or brackets that are generally similar to the upper clamps **1816** discussed above with reference to FIGS. **18**A and **18**B.

(145) As best shown in FIG. **34**C, each of the stowage brackets **3436***a*-**3436***c* can include a corresponding slot or notch **3432***a*-**3432***c*, respectively, that is configured to receive a portion of the transfer ramp **1802** as the transfer ramp **1802** is retracted to the stowed position. The notches **3432***a*-**3432***c* can each include a bumper **3435** made of rubber, plastic, or another suitable material. Additionally, or alternatively, the notches **3432***a*-**3432***c* can each include a pie- or wedge-shaped opening or slot that is configured to receive the transfer ramp **1802**. For example, each of the notches **3432***a*-**3432***c* can include a first ramp portion **3434***a* (or first tapered surface) and/or a second ramp portion **3434***b* (or second tapered surface) that are configured to guide the transfer ramp **1802** into the corresponding one of the notches **3432***a*-**3432***c* as the transfer ramp **1802** is retracted toward the stowed position. Continuing with this example, as the transfer ramp **1802** is retracted to return the system **1800** to the stowed position, the transfer ramp **1802** can be (e.g., tightly) seated within the notches **3432***a*-**3432***c* in the stowage brackets **3436***a*-**3436***c*, respectively, of the stowage stop. Thus, the stowage stop is expected to (a) stow the transfer ramp **1802** in a generally flat or horizontal orientation, (b) reduce, prevent, or hinder excessive vibration/movement of the transfer ramp **1802** and/or the system **1800** while the vehicle/trailer is being driven/towed, and/or (c) limit, reduce, or minimize wear and tear on the transfer ramp **1802** and/or one or more of the actuators **1850***a*-**1850***c* that may otherwise result from the excessive vibration/movement.

(146) In the embodiment illustrated in FIGS. **34**A-**34**C, the stowage stop includes three stowage brackets **3436***a*-**3436***c* with corresponding notches **3432***a*-**3432***c*, respectively. As discussed above, the transfer ramp **1802** can be generally flexible (e.g., to address misalignment between the top platform **1822** and a top surface **232** of another structure **230** (FIG. **22**N)). Therefore, the stowage brackets **3436***a*-**3436***c* can be laterally spaced apart from one another to support the sides and

middle of the transfer ramp **1802** at least while the transfer ramp **1802** is seated within the notches **3432***a*-**3432***c* (e.g., to limit, prevent, or reduce sagging of the sides and/or middle of the transfer ramp **1802**). In other embodiments, a different number of stowage brackets **3436** (e.g., one, two, or more than three stowage brackets **3436**) can be used. As a specific example, the second stowage bracket **3436***b* can be omitted in some embodiments. Continuing with this example, the distance between the first stowage bracket **3436***a* and the second stowage bracket **3436***c* can be reduced.

(147) In some embodiments, the transfer ramp **1802** can be generally flexible in a direction generally along or parallel to the longitudinal axis of the vehicle/trailer. In these embodiments, the depths of the notches **3432***a*-**3432***c* can be configured to receive enough of the transfer ramp **1802** to support the transfer ramp **1802** in the direction generally along or parallel to the longitudinal axis of the vehicle/trailer (e.g., to prevent, limit, or reduce sagging of the transfer ramp **1802** in this direction at least while the transfer ramp **1802** is seated within the notches **1832***a*-**1832***c*). For example, as best shown in FIG. **22**N, the notches **1832***a*-**1832***c* can be configured to receive the transfer ramp **1802** such that the transfer ramp **1802**, when fully seated in the notches **1832***a*-**1832***c*, is supported by the stowage brackets **3436***a*-**3436***c* to a point **2249** or axis around which the transfer ramp **1802** is configured to rotate when released from the ramp latching system **1810**. As specific examples, the notches **1832***a*-**1832***c* can be configured to receive the transfer ramp **1802** such that the transfer ramp **1802**, when fully seated in the notches **1832***a*-**1832***c*, is supported by the stowage brackets **3436***a*-**3436***c* beyond a midpoint of the transfer ramp **1802** and/or to points located beyond the ends of the extendable arms **1804***a* and **1804***b*, such as to points beyond or along the ramp latching system **1810**.

(148) Although the steps of the methods described herein (e.g., with reference to FIGS. **22**A-**24**N and/or **34**A-**34**C) are discussed and/or illustrated in a particular order, the methods described herein are not so limited. In other embodiments, the methods described herein can be performed in different orders. In these and other embodiments, any of the steps of the methods described herein can be performed before, during, and/or after any of the other steps of the methods described herein. Furthermore, a person skilled in the art will readily recognize that the methods described herein can be altered and still remain within these and other embodiments of the present technology. For example, one or more steps of the methods described herein can be omitted and/or repeated in some embodiments.

(149) FIGS. **25**A-**25**D are perspective views of a transfer ramp stowage system **2515** configured in accordance with various embodiments of the present technology. As shown, the transfer ramp stowage system **2515** includes a ramp latching system **2510** and a ramp tilting system **2575**. The ramp latching system **2510** can be generally similar to the ramp latching system **110** described above with reference to the system **100** of FIGS. **1**-**8**L. For example, the ramp latching system **2510** can include a cable release latch **2511** (e.g., similar to an automotive hood latch or another suitable type of latch) that is configured to releasably hold a latchable portion **2509** of the transfer ramp **1802**. The latchable portion **2509** can be released from the cable release latch **2511** via a cable **2512** and a corresponding actuation mechanism (not shown). As shown, the ramp tilting system **2575** includes a cable **2582** and a pulley system **2580**. The cable **2582** is attached at one end to the transfer ramp **1802**, is routed through a loop of the pulley system **2580** and/or across a pulley of the pulley system **2580**, and is attached at another end to a portion (e.g., a mounting bracket **1806**, an extendable arm **1804**) of the system **1800**, to the top platform **1822**, to the liftgate **1820**, and/or to the vehicle/trailer.

(150) In operation, the ramp latching system **2510** is configured to hold the transfer ramp **1802** in the stowed orientation. More specifically, the ramp latching system **2510** releasably holds the latchable portion **2509** of the transfer ramp **1802** in the cable release latch **2511** until the cable **2512** and the corresponding actuation mechanism (not shown) is used to release the cable release latch **2511**. Thereafter, the transfer ramp **1802** is permitted to pivot or rotate from the stowed orientation to a generally flat or horizontal orientation shown in FIGS. **25**A and **25**B. To return the transfer

ramp **1802** to the stowed orientation, the first and second extendable arms **1804***a* and **1804***b* can be extended in a direction generally along or parallel to arrow A (FIG. **25**B) such that the cable **2582** is brought taught against the pulley system **2580** (as shown in FIG. **25**C). Continued extension of the first and second extendable arms **1804***a* and **1804***b* in the direction generally along or parallel to the arrow A exerts a force on the transfer ramp **1802** via the cable **2582** that pivots or rotates the transfer ramp **1802** in a direction generally along or parallel to arrow B shown in FIG. **25**C (e.g., until the transfer ramp **1802** is returned to the stowed orientation in which the latchable portion **2509** transfer ramp **1802** is engaged with and is held by the cable release latch **2511** of the ramp latching system **2510**, as shown in FIG. **25**D).

(151) In some embodiments, the transfer ramp stowage system **2515** can include a structure (not shown) that houses or is otherwise configured to protect the cable **2582** and/or the pulley system **2580** of the ramp tilting system **2575**. For example, the structure can include a side bracket that includes a channel or groove through which the cable **2582** can be routed between the transfer ramp **1802** and the portion of the system **1800**, the top platform **1822**, the liftgate **1820**, and/or the vehicle/trailer attached to an opposite end of the cable **2582**). Housing the cable **2582**, the pulley system **2580**, and/or other components of the ramp tilting system **2575** in the structure can protect these components from damage, such as from a pallet, a pallet jack, a user, the liftgate **1820**, and/or another object.

(152) FIGS. **26**A-**26**E are partially schematic perspective views of another transfer ramp stowage system **2615** configured in accordance with various embodiments of the present technology. As shown, the transfer ramp stowage system **2615** includes a ramp latching system **2610** and a ramp tilting system **2675**. The ramp latching system **2610** is generally similar to the ramp latching system **110** described above with reference to the system **100** of FIGS. **1-8**L. For example, the ramp latching system **2610** includes a cable release latch **2611** (e.g., similar to an automotive hood latch or another suitable type of latch) that is configured to releasably hold a latchable portion **2609** of the transfer ramp **1802**. The latchable portion **2609** can be released from the cable release latch **2611** via a cable (not shown) and a corresponding actuation mechanism (not shown). As shown, the ramp tilting system **2675** includes a cable **2682** and a routing mechanism **2680** (e.g., a loop, slot, pulley, and/or another feature). The cable **2682** is attached at one end to the transfer ramp **1802** at a location **2685**, is routed across or through the routing mechanism **2680**, and is attached at another end to a portion (e.g., a mounting bracket **1806**, an extendable arm **1804**) of the system **1800**, to the liftgate **1820**, and/or to the vehicle/trailer.

(153) In operation, the ramp latching system **2610** is configured to hold the transfer ramp **1802** in the stowed orientation. More specifically, the ramp latching system **2610** releasably holds the latchable portion **2609** of the transfer ramp **1802** in the cable release latch **2611** until the cable (not shown) and the corresponding actuation mechanism (not shown) is used to release the cable release latch **2511**. Thereafter, the transfer ramp **1802** is permitted to pivot or rotate from the stowed orientation to a generally flat or horizontal orientation shown in FIGS. **26**A-**26**C. To return the transfer ramp **1802** to the stowed orientation, the first and second extendable arms **1804***a* and **1804***b* can be extended in a direction generally along or parallel to arrow B (FIG. **26**C) such that the cable **2682** is brought taught against the routing mechanism **2680** (as shown in FIG. **26**C). Continued extension of the first and second extendable arms **1804***a* and **1804***b* in the direction generally along or parallel to the arrow B exerts a force on the transfer ramp **1802** via the cable **2682** and in a direction generally along or parallel to arrow C (FIG. **26**C) that pivots or rotates the transfer ramp **1802** in a direction generally along or parallel to arrow D shown in FIG. **26**C (e.g., until the transfer ramp **1802** is returned to the stowed orientation in which the latchable portion **2609** transfer ramp **1802** is engaged with and is held by the cable release latch **2611** of the ramp latching system **2610**, as shown in FIGS. **26**D and **26**E).

(154) In some embodiments, the transfer ramp stowage system **2615** can include a structure (not shown) that houses or is otherwise configured to protect the cable **2682** and/or the routing

mechanism **2680** of the ramp tilting system **2675**. For example, the structure can include a side bracket that includes a channel or groove through which the cable **2682** can be routed between the transfer ramp **1802** and the portion of the system **1800**, the top platform **1822**, the liftgate **1820**, and/or the vehicle/trailer attached to an opposite end of the cable **2682**). Housing the cable **2682**, the routing mechanism **2680**, and/or other components of the ramp tilting system **2675** in the structure can protect these components from damage, such as from a pallet, a pallet jack, a user, the liftgate **1820**, and/or another object.

(155) Referring to FIG. **26**B, in some embodiments, the cable **2682** can be pulled in a direction generally along or parallel to arrow A. For example, while the transfer ramp **1802** is resting on the top platform **1822** of the liftgate **1820** and/or on the top surface **232** of the other structure **230** (FIGS. **22**A-**22**N), the cable **2682** can be pulled in the direction generally along or parallel to the arrow A. In turn, a front of the transfer ramp **1802** can be pulled upwards in a direction generally along or parallel to the arrow C (FIG. **26**C) and thereby exert a downwards force against the back of the transfer ramp **1802** (e.g., to better flatten the transfer ramp **1802** against the top platform **1822**).

(156) FIG. **27** is a partially schematic perspective view of a curved track **2704** configured in accordance with various embodiments of the present technology. The curved track **2704** can be incorporated into and/or interfaced with the first and second extendable arms **1804***a* and **1804***b* in some embodiments, such as to navigate the transfer ramp **1802** through a gap that exists between the top platform **1822** of the liftgate **1820** and a top surface **232** of another structure **230**. In some embodiments, the curved track **2704** can be used in addition to or in lieu of the third actuator **1850***c* and/or fourth actuators **1850***d* (FIGS. **18**A, **18**B, and **24**A).

(157) FIG. **28** is a partially schematic perspective view of a user controls arrangement **2815** configured in accordance with various embodiments of the present technology. As shown, the user controls arrangement **2815** includes a first switch **2816**, a second switch **2817**, and a visual indicator **2818**. In the illustrated embodiment, the first switch **2816** is a momentary switch or a toggle switch, and the second switch **2817** is a button switch. The first switch **2816** and/or the second switch **2817** can be another suitable type of switch in other embodiments of the present technology. The user controls arrangement **2815** further includes a housing **2812** configured to house electronics (not shown) of the system, such as an electronic control unit (ECU) or other controller, microcontroller, processor, integrated circuit, memory, etc. The visual indicator **2818** can be used to notify a user/operator of an error (e.g., with the system; with the configuration of the vehicle/trailer relative to another vehicle, trailer, loading dock, or other structure; etc.) and/or to convey other information to the user/operator.

(158) In some embodiments, the first switch **2816** can be used to extend and/or retract a corresponding transfer ramp system. For example, when an operator actuates the first switch **2816** (e.g., by pressing the first switch **2816** in a first direction, such as up), the user controls arrangement **2815** can cause the transfer ramp system to deploy or extend the transfer ramp, such as from beneath the rear platform of the truck or trailer. Extension of the transfer ramp can continue for as long as the operator keeps the first switch **2816** actuated and/or until the transfer ramp is positioned in a desired position (e.g., a fully extended position). As another example, when an operator actuates the first switch **2816** (e.g., by pressing the first switch **2816** in a second direction, which can be the same direction as or a different direction from the first direction, such as down), the user controls arrangement **2815** can cause the transfer ramp system to retract the transfer ramp toward its stowed position beneath the rear platform of the truck or trailer. Retraction of the transfer ramp can continue for as long as the operator keeps the first switch **2816** actuated and/or until the transfer ramp is positioned in a desired position (e.g., a fully stowed position).

(159) The second switch **2817** can be used to lower the transfer ramp and/or press the transfer ramp down into (a) the rear platform of the truck or trailer that includes the transfer ramp system and/or (b) a loading dock platform, a rear platform of another truck or trailer, and/or another platform. In

some embodiments, the second switch **2817** can be used to lower the transfer ramp after (e.g., only after) the transfer ramp has been moved to a desired position (e.g., a fully deployed position) and/or the transfer ramp has been pivoted such that it is oriented generally parallel to the rear platform (e.g., by rotating the transfer ramp to the generally parallel orientation after releasing a mechanical latch and/or using an electric motor or other actuation mechanism). The press down process can continue for as long as the operator keeps the button switch actuated and/or until the transfer ramp system determines (e.g., by monitoring current, voltage, or another measured indicator of the transfer ramp system) that the transfer ramp has been sufficiently pressed down against the platforms to facilitate transferring cargo across the transfer ramp.

(160) An example operation flow of the user controls arrangement **2815** is as follows: (1) an operator actuates the first switch **2816** to deploy the transfer ramp from beneath the rear platform of a truck or trailer; (2) after the transfer ramp reaches a fully deployed position or a position located above the rear platform and another platform, the operator de-actuates the first switch **2816** (or the first switch de-actuates automatically); (3) the transfer ramp is released (e.g., by operating a mechanical latch, such as the mechanical latch discussed in greater detail below with reference to FIGS. **29**A and **29**B) and is pivoted (e.g., freely, by the operator, using an electric motor or other actuator that controls an angle of the transfer ramp relative to the extendable arms, using another mechanism) to an orientation in which the transfer ramp is generally parallel to the platforms; (4) an operator actuates the second switch **2817** to lower the transfer ramp towards and/or against the platforms; (5) the system continues to lower the transfer ramp against the platforms beyond the point in time at which the transfer ramp initially makes contact with the platforms to level the platforms to a same height (or more similar heights) and/or warp the transfer ramp to conform the transfer ramp to the arrangement of the platforms/make the transfer ramp as flat as possible; (6) this lowering process can continue until the operator de-actuators the second switch **2817** and/or the system determines that the transfer ramp has been sufficiently lowered against the platforms; (7) cargo is transferred across the transfer ramp; (8) the operator actuates the first switch **2816** to extend/raise the platform off of the platforms; (9) the transfer ramp is pivoted (e.g., by an operator, an electric motor, a transfer ramp stowage system, or another mechanism) back into alignment with the extendable arms of the system; and (10) the operator actuates the first switch **2816** to retract/lower the transfer ramp back to its stowed position beneath the rear platform of the truck or trailer.

(161) FIGS. **29**A and **29**B are a partially schematic side perspective views of the transfer ramp system **1800**. As shown, the transfer ramp system **1800** includes a ramp latching system **1810**. The ramp latching system **1810** can include a lever/pin and corresponding catches. More specifically, the ramp latching system **1810** in the illustrated embodiments includes a mechanical latch **2911**, a first notch **2914***a* or catch, and a second notch **2914***b* or catch. As best shown in FIG. **29**B, the first notch **2914***a* and the second notch **2914***b* can be formed at an end portion of the second extendable arm **1804***b*. In other embodiments, the first notch **2914***a* and/or the second notch **2914***b* can be formed in other components of the system **1800** and/or can be located at other positions than shown in FIG. **29**B.

(162) In the illustrated embodiment, the mechanical latch **2911** is a thumb latch that, when manually pressed or actuated by a user/operator, pivots a latchable portion **2909** (or lever) of the mechanical latch **2911** away from and/or out of the first notch **2914***a* and/or the second notch **2914***b* such that the transfer ramp **1802** can be pivoted (e.g., freely, manually by the user/operator, using an electric motor, using struts, using another mechanism) about a pivot point **2249**/pivot axis between (a) the stowed position/orientation shown in FIG. **29**A (in which the transfer ramp **1802** is generally in line with the extendable arm **1804***b*) and (*b*) the generally flat, horizontal, or level position/orientation shown in FIG. **29**B in which the transfer ramp **1802** is generally parallel to the top platform **1822** of the truck/trailer that includes the transfer ramp system **1800**.

(163) In the illustrated embodiment, the first notch **2914***a* corresponds to the stowed

position/orientation. In particular, after the mechanical latch **2911** is actuated, the transfer ramp **1802** is rotated, and/or the mechanical latch **2911** is subsequently released, the latchable portion **2909** can be swung toward and/or into the first notch **2914***a*. While the latchable portion **2909** is positioned within the first notch **2914***a*, the first notch **2914***a* can (i) hold the transfer ramp **1802** in the stowed position/orientation and/or (ii) prevent or hinder the transfer ramp **1802** from rotating about the pivot point **2249**.
(164) The second notch **2914***b* can correspond to the generally flat, horizontal, or level position/orientation. In particular, after the mechanical latch **2911** is actuated, the transfer ramp **1802** is rotated, and/or the mechanical latch **2911** is subsequently released, the latchable portion **2909** can be swung toward and/or into the second notch **2914***b*. While the latchable portion **2909** is positioned within the second notch **2914***b*, the second notch **2914***b* can (i) hold the transfer ramp in to the generally flat, horizonal, or level position/orientation and/or (ii) prevent or hinder the transfer ramp **1802** form rotating about the pivot point **2249**.
(165) Although the ramp latching system **1810** illustrated in FIGS. **29**A and **29**B includes a mechanical thumb latch with corresponding notch catches, the ramp latching system can employ other suitable latches/pins/fasteners and/or corresponding catches/locks in other embodiments of the present technology. For example, a cotter pin or a ball lock pin can be used in addition to or in lieu of the thumb latch shown in FIGS. **29**A and **29**B. Continuing with these examples, the cotter pin or the ball lock pin can be used to lock the transfer ramp **1802** in the stowed position/orientation and/or the generally flat, horizontal, or level position/orientation, such as using corresponding locking/catching mechanisms (e.g., slots, notches, and/or apertures) that are configured to reversibly receive/retain the cotter pin or the ball lock pin. As another example, a push button lock and corresponding locking/catching mechanism (e.g., apertures) can be used to transition the transfer ramp **1802** and/or hold the transfer ramp **1802** in the stowed position/orientation and/or the generally flat, horizontal, or level position/orientation.
(166) FIG. **30** is a partially schematic diagram of a controls system **3000** configured in accordance with various embodiments of the present technology. As shown, the controls system **3000** includes an electronic control unit **3010** ("ECU **3010**") or microcontroller, a user controls arrangement **3015** (also referred to herein as "user buttons" or a "user interface"), and a plurality of actuators **3050***a*-**3050***d*. Actuators **3050***a* and **3050***b* can be extension actuators (e.g., the extension actuators **1850***a* and **1850***b* discussed in detail above with reference to FIGS. **18**A-**24**N), and actuators **3050***c* and **3050***d* can be rotation actuators (e.g., the rotation actuators **1850***c* and **1850***d* discussed in detail above with reference to FIGS. **18**A-**24**N). In some embodiments, the controls system **3000** can be used to actuate, control, move, calibrate, etc. a transfer ramp system, such as one of more of the transfer ramp systems described herein. Indeed, all or a portion of the controls system **3000** can be included in (or be components of) one or more of the transfer ramp systems described herein.
(167) As shown in FIG. **30**, the user controls arrangement **3015** can be used to control the ECU **3010**, which in turn can control the actuators **3050***a*-**3050***d*. For example, an operator can actuate (e.g., a switch of) the user controls arrangement **3015** to awaken the ECU **3010** and/or instruct the controls system **300** to perform an action. In turn, the ECU **3010** can interpret user input received via the user controls arrangement **3015**, and can control the actuators **3050***a*-**3050***d* accordingly. For example, when an operator uses the user controls arrangement **3015** to instruct the system to extend/raise or retract/lower a transfer ramp, the ECU **3040** can interpret this input and actuate the actuators **3050***a* and **3050***b* to extend/raise/retract/lower the transfer ramp (e.g., via extension/retraction of extendible arms of the system) accordingly. In the event that movement of the transfer ramp requires rotation of the transfer ramp (e.g., to navigate through a gap between platforms), the ECU **3010** can additionally, or alternatively, actuate the actuators **3050***c* and **3050***d* to tilt the extendible arms and/or rotate the transfer ramp accordingly. As another example, when an operator uses the user controls arrangement **3015** to instruct the system to lower the transfer ramp towards or press the transfer ramp down against platforms, the ECU **3010** can interpret this input

and actuate one or more of the actuators **3050***a*-**3050***d* accordingly.
(168) In some embodiments, the ECU **3010** can monitor the absolute positions of the actuators and/or perform a speed control function to ensure that the transfer ramp is deployed/retracted smoothly and/or without damaging the transfer ramp system, the truck/trailer, and/or another object. For example, as the ECU **3010** actuates the extension actuators **3050***a* and **3050***b* to extend a transfer ramp, the ECU **3010** can monitor the position of the actuators **3050***a* and **3050***b* (e.g., via potentiometers corresponding to the actuators **3050***a* and **3050***b*) to ensure that they are extending (e.g., deploying) the transfer ramp at the same rate and/or along a desired or predetermined path (e.g., the S-shaped path shown in FIG. **21**). In the event that the ECU **3010** determines that one of the actuators **3050***a* and **3050***b* is extending the transfer ramp at a faster rate than the other of the actuators **3050***a* and **3050***b* (e.g., because the other of the actuators **3050***a* and **3050***b* is fouled up with dirt or other debris) and/or in the event that the ECU **3010** determines that the transfer ramp has veered/deviated from the desired/predetermined path, the ECU **3010** can implement one or more functions (e.g., a speed control function, a position control function, a current control function, pulse width modulation, gradient descent algorithms, etc.) to (a) reduce the speed with which the one of the actuators **3050***a* and **3050***b* is extending the transfer ramp to match or better align with the speed with which the other of the actuators **3050***a* and **3050***b* is extending the transfer ramp and/or (b) activate one or more of the actuators **3050***a*-**3050***d* to return the transfer ramp to the desired/predetermined path.
(169) As also shown in FIG. **30**, the controls system **3000** can additionally include a diagnostics board/module **3020** in some embodiments. An example of the diagnostics board/module **3020** is shown in FIG. **32** and described in greater detail below.
(170) FIGS. **31**A and **31**B illustrate a flow diagram showing a process **3100** for controlling a transfer ramp system in accordance with various embodiments of the present technology. Referring first to FIG. **31**A, the process **3100** can begin by receiving operator input (block **3101**), such as via a button/switch of a user controls arrangement. This input can awaken and/or power up an ECU of the transfer ramp system (block **3102**), and the ECU can identify (block **3103**) what user input was received (e.g., by determining what button of the user controls arrangement was pressed). At block **3104**, the ECU can select a trajectory (e.g., a predetermined or desired path) for the transfer ramp based on the user input, and can retrieve (block **3105**) the corresponding trajectory information from memory of the ECU. As shown, the memory of the ECU can include a transfer ramp deploy trajectory **3105***a*, a transfer ramp stow trajectory **3105***b*, a transfer ramp press-down trajectory **3105***c*, and/or a transfer ramp lift-up trajectory **3105***d*. All or a subset of these trajectories **3105***a*-**3105***d* can be preset, predetermined, or precalculated. The memory of the ECU is also shown as including information **3105***e* for calibrating the actuator potentiometers of the system, which the ECU can use at block **3106** when reading potentiometer values and converting them to actuator lengths to determine (e.g., absolute) positions of the actuators. Calibration of the actuator potentiometers is discussed in greater detail below with reference to FIGS. **32** and **33**.
(171) Referring now to FIG. **31**B, the process **3100** can continue at block **3107** by the ECU determining a position of the transfer ramp using forward kinematics. Thereafter, the process **3100** can enter a model prediction control (MPC) portion **3108** of the process **3100**. More specifically, at block **3109**, a future position of the transfer ramp can be simulated given a set of control commands. At block **3110**, a cost function can be calculated based on the proximity of the future position of the transfer deck to the trajectory. This step can be based at least in part on the trajectory information selected/retrieved from the ECU memory at block **3104** (FIG. **31**A). At block **3111**, a derivative of the cost function can be calculated relative to the control commands to obtain a gradient. At block **3112**, a gradient descent algorithm (e.g., using machine learning) can be performed. At block **3113**, control commands can be generated/updated. At block **3114**, the generated/updated control commands can be used to pulse-width modulate actuators of the system. Thereafter, the process **3100** can return to block **3103** (FIG. **31**A) and/or terminate (e.g., the ECU

can go to sleep).

(172) FIG. **32** is a partially schematic diagram of a diagnostics board **3220** or module configured in accordance with various embodiments of the present technology. The diagnostics board **3220** can be used by technicians to calibrate and/or service a transfer ramp system of the present technology. As shown, the diagnostics board **3220** includes (a) a button or switch **3221** for connecting to and/or activating (e.g., remotely activating) a transfer ramp system of the present technology; (b) a JOG controller **3222** for extending/retracting/raising/lower the extendable arms and/or transfer ramp of the transfer ramp system via actuation of one or more actuators of the system; (c) a button or switch **3223** for actuating one or more actuators of the system to extend extendible arms and/or a transfer ramp of the transfer ramp system; (d) a button or switch **3224** for actuating one or more actuators of the system to retract extendible arms and/or a transfer ramp of the transfer ramp system; and (e) a button or switch **3225** for actuating one or more actuators of the system to lower extendible arms and/or a transfer ramp (e.g., as part of the press down process described in greater detail above).

(173) In the illustrated embodiment, the diagnostics board **3220** additionally includes a screen or display **3227** (e.g., for conveying information to a user or technician). For example, the display **3227** can be used to indicate current (e.g., absolute) positions of potentiometers that are used to determine a current position of corresponding actuators of the system. The diagnostics board **3220** can additionally include a calibration button **3228**, such as for calibrating the potentiometers of the transfer ramp system. Additionally, or alternatively, the diagnostics board **3220** can includes a speed control mechanism **3229**. Adjusting the position of the speed control mechanism **3229** (e.g., by sliding the speed control mechanism **3229** up or down along the edge of the diagnostics board **3220**) can adjust the speed with which the actuators of the transfer control system extend/retract the extendible arms and/or a transfer ramp of the transfer ramp system. For example, sliding the speed control mechanism **3229** upwards toward the top of the diagnostics board **3220** can increase the extension/retraction speed, and sliding the speed control mechanism **3229** downwards toward the bottom of the diagnostics board can decrease the extension/retraction speed.

(174) FIG. **33** is a partially schematic perspective view of a travel limiter **3358** installed on the second actuator **1850***b* of the transfer ramp system **1800**. In some embodiments, the travel limiter **3358** can be used to calibrate a potentiometer corresponding to the second actuator **1850***b*. More specifically, the travel limiter **3358** can be positioned at an end portion of a rod **3354** of the second actuator **1850***b*. To calibrate the potentiometer corresponding to the second actuator **1850***b*, the rod **3354** can be retracted into a body **3356** of the second actuator **1850***b* until the travel limiter **3358** abuts against a front face of the body **3356**. An ECU of the system **1800** can utilize this fully retracted position as a “zero” point for calibration-purposes. For example, the ECU can (e.g., when a calibration button, such as the calibration button **3228** on the diagnostics board **3220** of FIG. **32**, is pressed) use the zero point and a corresponding reading on the potentiometer to determine a fixed length of the second actuator **1850***b*. This information can be used to determine an absolute position of (e.g., the travel limiter **3358** of) the second actuator **1850***b* at any given time, which the ECU can use to extend and/or retract the transfer ramp **1802** of the system **1800** (e.g., along a predetermined path). As shown in FIG. **33**, the travel limiter **3358** of the second actuator **1850***b* can include a chamfer **3359**, such as to avoid contacting the liftgate **1820**. In some embodiments, the first actuator **1850***a* of the system **1800** can include a travel limiter that is generally similar to the travel limiter **3358** and/or that can be used to calibrate a potentiometer corresponding to the first actuator **1850***a*. Additionally, or alternatively, as best shown in FIGS. **19**A and **23**C, the third and/or fourth actuators **1850***c* and **1850***d* can include travel limiters **1958** that are generally similar to the travel limiter **3358** and/or that can be used to calibrate potentiometers corresponding to the third and/or fourth actuators **1850***c* and **1850***d*.

(175) Although not shown so as to avoid unnecessarily obscuring the description of the embodiments of the technology, any of the forgoing systems and methods described above can include and/or be performed by a computing device configured to direct and/or arrange

components of the systems and/or to receive, arrange, store, analyze, and/or otherwise process data received, for example, from the machine and/or other components of the systems. As such, such a computing device includes the necessary hardware and corresponding computer-executable instructions to perform these tasks. More specifically, a computing device configured in accordance with an embodiment of the present technology can include a processor, a storage device, input/output device, one or more sensors, and/or any other suitable subsystems and/or components (e.g., displays, speakers, communication modules, etc.). The storage device can include a set of circuits or a network of storage components configured to retain information and provide access to the retained information. For example, the storage device can include volatile and/or non-volatile memory. As a more specific example, the storage device can include random access memory (RAM), magnetic disks or tapes, and/or flash memory.

(176) The computing device can also include (e.g., non-transitory) computer readable media (e.g., the storage device, disk drives, and/or other storage media) including computer-executable instructions stored thereon that, when executed by the processor and/or computing device, cause the systems to perform one or more of the methods described herein. Moreover, the processor can be configured for performing or otherwise controlling steps, calculations, analysis, and any other functions associated with the methods described herein.

(177) In some embodiments, the storage device can store one or more databases used to store data collected by the systems as well as data used to direct and/or adjust components of the systems. In one embodiment, for example, a database is an HTML file designed by the assignee of the present disclosure. In other embodiments, however, data is stored in other types of databases or data files.

(178) One of ordinary skill in the art will understand that various components of the systems (e.g., the computing device) can be further divided into subcomponents, or that various components and functions of the systems may be combined and integrated. In addition, these components can communicate via wired and/or wireless communication, as well as by information contained in the storage media.

C. Examples

(179) Several aspects of the present technology are set forth in the following examples. Although several aspects of the present technology are set forth in examples directed to systems and methods, these aspects of the present technology can similarly be set forth in examples directed to methods and systems, respectively, in other embodiments. Additionally, these aspects of the present technology may be set forth in examples directed to devices and/or (e.g., non-transitory) computer-readable media in other embodiments. 1. A transfer ramp system mountable beneath a top platform at a rear of a vehicle or trailer, the transfer ramp system comprising: a transfer ramp; and at least one extendable arm attached to the transfer ramp and usable to extend or retract the transfer ramp, wherein, when the transfer ramp system is mounted beneath the top platform, the transfer ramp is moveable between (a) a stowed position in which the transfer ramp is positioned beneath the top platform and (b) a deployed position in which the transfer ramp is usable to bridge a gap between the top platform and another structure for moving cargo between (i) the vehicle or trailer and (ii) the other structure. 2. The transfer ramp system of example 1 wherein the transfer ramp system is mountable beneath the top platform at the rear of the vehicle or trailer such that, when the transfer ramp is moved between the stowed position and the deployed position, the transfer ramp is navigated through a space below the top platform and above a liftgate of the vehicle or trailer. 3. The transfer ramp system of example 1 or example 2 wherein the at least one extendable arm includes a first extendable arm coupled to a first side of the transfer ramp and a second extendable arm coupled to a second side of the transfer ramp opposite the first side. 4. The transfer ramp system of any of examples 1-3, further comprising at least one actuator configured to extend or retract the at least one extendable arm to extend or retract, respectively, the transfer ramp. 5. The transfer ramp system of example 4 wherein the at least one actuator includes a linear actuator. 6. The transfer ramp system of example 4 or example 5 wherein the at least one actuator includes an

electric linear actuator. 7. The transfer ramp system of any of examples 4-6 wherein the at least one actuator includes a pneumatic linear actuator. 8. The transfer ramp system of any of examples 4-7 wherein the at least one actuator includes a first actuator usable to extend or retract the at least one extendable arm, and wherein the transfer ramp system further comprises a second actuator usable to rotate the at least one extendable arm. 9. The transfer ramp system of any of examples 4-8 wherein: the at least one extendable arm includes a first extendable arm extending generally along a first longitudinal axis and a second extendable arm different from the first extendable arm, wherein the second extendable arm extends generally along a second longitudinal axis; the at least one actuator includes a first actuator and a second actuator; the transfer ramp system further comprises a third actuator and a fourth actuator; the first actuator is usable to extend or retract the first extendable arm generally along the first longitudinal axis; the third actuator is usable to rotate the first extendable arm and adjust a pitch of the first extendable arm; the second actuator is usable to extend or retract the second extendable arm generally along the second longitudinal axis; and the fourth actuator is usable to rotate the second extendable arm and adjust a pitch of the second extendable arm. 10. The transfer ramp system of any of examples 1-9, further comprising: at least one actuator configured to manipulate the at least one extendable arm; and a microcontroller configured to control the at least one actuator to move the transfer ramp between the stowed position and the deployed position. 11. The transfer ramp system of example 10 wherein the microcontroller is configured to control the at least one actuator to move the transfer ramp between the stowed position and the deployed position generally along a predetermined path. 12. The transfer ramp system of example 10 or example 11 wherein the microcontroller is configured to, based at least in part on feedback received from the at least one actuator, control the at least one actuator to move the transfer ramp (i) between the stowed position and the deployed position and/or (ii) along a predetermined path. 13. The transfer ramp system of example 11 or example 12 wherein the predetermined path is generally S-shaped such that, as the transfer ramp is moved between the stowed position and the deployed position, the transfer ramp is navigated under the top platform, through the gap between the top platform and the other structure, and above the top platform and the other structure. 14. The transfer ramp system of any of examples 1-13 wherein the transfer ramp is configured to rotate between (i) a stowed orientation for stowage beneath the top platform and (ii) a deployed orientation in which a top surface of the transfer ramp is generally parallel to a top surface of the top platform. 15. The transfer ramp system of example 14 wherein, at least while the transfer ramp is in the stowed orientation, the top surface of the transfer ramp is generally parallel to a longitudinal axis of an extendable arm of the at least one extendable arm. 16. The transfer ramp system of example 14 or example 15, further comprising at least one strut having a first end attached to an extendable arm of the at least one extendable arm and a second end attached to the transfer ramp, wherein the at least one strut is usable to pivot the transfer ramp at least partway between (i) the stowed orientation and (ii) the deployed orientation. 17. The transfer ramp system of example 16 wherein the at least one strut includes a gas strut or an air cylinder. 18. The transfer ramp system of any of examples 14-17, further comprising at least one spring configured to pivot the transfer ramp at least partway between (i) the stowed orientation and (ii) the deployed orientation. 19. The transfer ramp system of example 18 wherein the at least one spring includes a torsion spring. 20. The transfer ramp system of any of examples 14-19, further comprising a ramp latching system configured to releasably hold the transfer ramp in the stowed orientation. 21. The transfer ramp system of example 20 wherein the ramp latching system includes a cable release latch. 22. The transfer ramp system of example 20 or example 21 wherein the ramp latching system is further configured to releasably hold the transfer ramp in the deployed orientation. 23. The transfer ramp system of any of examples 14-22, further comprising a ramp latching system, wherein the ramp latching system includes: a first latch configured to releasably hold the transfer ramp in the stowed orientation; or a second latch configured to releasably hold the transfer ramp in the deployed orientation. 24. The transfer ramp system of example 23 wherein the ramp latching system includes the first latch and the second latch. 25. The transfer ramp system of example 23 or example 24 wherein the first latch, the second latch, or a combination thereof includes (a) a cotter pin, a ball lock pin, or a mechanical thumb latch having a corresponding lever; and (b) a corresponding catch. 26. The transfer ramp system of any of examples 23-25 wherein the first latch, the second latch, or a combination thereof are configured to be manually actuated by an operator of the transfer ramp system. 27. The transfer ramp system of any of examples 23-26 wherein actuation of the first latch, the second latch, or a combination thereof is automated and controlled by the transfer ramp system. 28. The transfer ramp system of any of examples 1-27, further comprising a ramp tilting system configured to rotate the transfer ramp at least partway between (i) a deployed position in which a top surface of the transfer ramp is generally parallel to the top platform and (ii) a stowed orientation for storage beneath the top platform. 29. The transfer ramp system of example 28 wherein the ramp tilting system includes a cable operably coupled to the transfer ramp such that, as the at least one extendable arm is extended while the transfer ramp is in the deployed position, a force is exerted on the transfer ramp via the cable to rotate the transfer ramp toward the stowed orientation. 30. The transfer ramp system of any of examples 1-29 wherein: when the transfer ramp system is mounted beneath the top platform at the rear of the vehicle or trailer, at least part of the transfer ramp system is positioned above a liftgate of the vehicle or trailer; and the transfer ramp system further comprises a bump stop to limit vertical motion of the liftgate toward the transfer ramp system. 31. The transfer ramp system of any of examples 1-30, further comprising a stowage stop configured to (a) limit retraction of the transfer ramp, (b) hold the transfer ramp at least while the transfer ramp is in the stowed position, or (c) a combination thereof. 32. The transfer ramp system of example 31 wherein the stowage stop includes one or more slots configured to receive the transfer ramp as the transfer ramp is moved toward the stowed position. 33. The transfer ramp system of any of examples 1-32 wherein the transfer ramp is generally flexible such that the transfer ramp is conformable to vertical misalignment between (a) opposite sides of the top platform, (b) opposite sides of a top surface of the other structure, (c) the top platform and the top surface, or (d) a combination thereof. 34. A cargo transferring system mountable to or under frame rails of a vehicle or trailer, the cargo transferring system comprising: a bridge deck; one or more extendable arms operably coupled to the bridge deck; and one or more actuators configured to extend and retract the one or more extendable arms; wherein, during operation of the cargo transferring system when mounted to or under the frame rails of the vehicle or trailer, the one or more actuators are controllable to transition the bridge deck between— a stowed position under the frame rails, and a deployed position in which the bridge deck is usable to transfer cargo between (i) a platform of the vehicle or trailer and (ii) a surface of another structure different from the vehicle or trailer. 35. The cargo transferring system of example 34 wherein the cargo transferring system is mountable to or under the frame rails of the vehicle or trailer such that, when the bridge deck is transitioned between the stowed position and the deployed position, the bridge deck is navigated through a space between the platform and a liftgate of the vehicle or trailer. 36. The cargo transferring system of example 34 or example 35 wherein: the one or more actuators are one or more first actuators configured to extend and retract the one or more extendable arms generally along one or more longitudinal axes of the one or more extendable arms; and the cargo transferring system further comprises one or more second actuators usable to adjust a pitch of the one or more extendable arms. 37. The cargo transferring system of any of examples 34-36 wherein the bridge deck is configured to rotate relative to the one or more extendable arms such that a top surface of the bridge deck is alignable with a top surface of the platform. 38. The cargo transferring system of any of examples 34-37, further comprising a ramp latching system configured to: releasably retain the bridge deck in a first orientation for stowage beneath the frame rails; and/or releasably retain the bridge deck in a second orientation in which a top surface of the bridge deck is generally parallel with a top surface of the platform. 39. The cargo transferring system of any of examples 34-38, further comprising a microcontroller configured to control movement of the bridge deck between the stowed position and the deployed position. 40. The cargo transferring system of example 39 wherein the microcontroller is configured to control the movement of the bridge deck such that the bridge deck is moved between the stowed position and the deployed position according to a predetermined path. 41. The cargo transferring system of example 39 or example 40 wherein the microcontroller is configured to, based at least in part on feedback received from the one or more actuators, control the movement of the bridge deck (i) between the stowed position and the deployed position and/or (ii) according to a predetermined path. 42. The cargo transferring system of example 40 or example 41 wherein the predetermined path is generally S-shaped and extends between a point beneath the frame rails and a point above the platform. 43. A transfer ramp system mountable to or under frame rails of a vehicle or trailer, the transfer ramp system comprising: a transfer ramp; one or more extendable arms operably coupled to the transfer ramp; one or more extension actuators operably coupled to the one or more extendable arms and configured to extend or retract the one or more extendable arms; and one or more rotation actuators operably coupled to the one or more extendable arms and configured to pivotably move the one or more extendable arms, wherein, when the transfer ramp system is mounted to or under the frame rails, the one or more extension actuators and the one or more rotation actuators are controllable to manipulate the one or more extendable arms to move the transfer ramp (i) from a position beneath a platform of the vehicle or trailer, (ii) through a gap between the platform and another structure physically separate from the vehicle or trailer, and (iii) to a position above or at least partially aligned with the platform and a top surface of the other structure such that the transfer ramp is usable to transfer cargo between the platform and the top surface. 44. The transfer ramp system of example 43 wherein: the one or more extendable arms include a first extendable arm extending generally along a first longitudinal axis and a second extendable arm different from the first extendable arm, wherein the second extendable arm extends generally along a second longitudinal axis; the one or more extension actuators include a first actuator and a second actuator; the one or more rotation actuators include a third actuator and a fourth actuator; the first actuator is configured to extend or retract the first extendable arm generally along the first longitudinal axis; the third actuator is configured, at least while the first actuator is stationary, pivot the first extendable arm and adjust a pitch of the first extendable arm; the second actuator is configured to extend or retract the second extendable arm generally along the second longitudinal axis; and the fourth actuator is configured, at least while the second actuator is stationary, to pivot the second extendable arm and adjust a pitch of the second extendable arm. 45. The transfer ramp system of example 43 or example 44, further comprising a microcontroller configured to control the one or more extension actuators and the one or more rotation actuators to manipulate the one or more extendable arms to move the transfer ramp (i) from the position beneath the platform, (ii) through the gap, and (iii) to the position above or at least partially aligned with the platform, according to a predetermined path. 46. The transfer ramp system of any of examples 43-45 wherein the transfer ramp is configured to pivotably move between (i) a stowed orientation for stowage beneath the platform and (ii) a deployed orientation in which a top surface of the transfer ramp is generally parallel to a top surface of the platform. 47. The transfer ramp system of any of examples 43-46, further comprising a ramp latching system configured to: releasably retain the transfer ramp in a first orientation for stowage beneath the platform; or releasably retain the transfer ramp in a second orientation in which a top surface of the transfer ramp is generally parallel with a top surface of the platform. 48. A method of operating a transfer ramp system mounted to or under frame rails of a vehicle or trailer, the method comprising: deploying a transfer ramp of the transfer ramp system from beneath the frame rails of the vehicle or trailer; transitioning the transfer ramp between (i) a stowed orientation corresponding an orientation of the transfer ramp when stowed under the frame rails and (ii) a deployed orientation in which a top surface of the transfer ramp is more aligned with a platform of the vehicle or trailer than the top surface of the transfer ramp in the stowed orientation; and positioning the transfer ramp such that the transfer ramp is usable to bridge a gap between the platform and a surface of another structure proximate the vehicle or trailer. 49. The method of example 48 wherein deploying the transfer ramp includes extending at least one extendable arm of the transfer ramp system that is operably coupled to the transfer ramp. 50. The method of example 49 wherein extending the at least one extendable arm includes actuating an actuator operably coupled to the at least one extendable arm. 51. The method of any of example 49 or example 50 wherein deploying the transfer ramp includes rotating the at least one extendable arm such that an angle between (i) a longitudinal axis of the at least one extendable arm and (ii) a top surface of the platform is increased, wherein rotating the at least one extendable arm includes actuating a second actuator operably coupled to the at least one extendable arm. 52. The method of any of examples 48-51 wherein deploying the transfer ramp includes navigating the transfer ramp through a space beneath the platform and above a liftgate of the vehicle or trailer. 53. The method of any of examples 48-52 wherein deploying the transfer ramp includes deploying the transfer ramp through the gap between the platform and the surface of the other structure. 54. The method of any of examples 48-53 wherein: the other structure is another vehicle, trailer, or loading dock; positioning the transfer ramp includes positioning the transfer ramp while the platform is positioned a distance from the surface of the other vehicle, trailer, or loading dock; and the distance corresponds to the gap and is less than or equal to a length of the transfer ramp. 55. The method of any of examples 48-54 wherein transitioning the transfer ramp includes extending a strut of the transfer ramp system that is operably coupled to the transfer ramp. 56. The method of example 55 wherein extending the strut includes filling the strut with gas or air. 57. The method of any of examples 48-56 wherein transitioning the transfer ramp includes releasing the transfer ramp from a latch. 58. The method of any of examples 48-57 wherein transitioning the transfer ramp includes pivoting the transfer ramp using a spring. 59. The method of any of examples 48-58 wherein positioning the transfer ramp includes positioning the transfer ramp such that at least a first portion of the transfer ramp rests on top of the surface of the other structure and such that at least a second portion of the transfer ramp rests on the platform. 60. The method of any of examples 48-59 wherein positioning the transfer ramp includes retracting an extendable arm of the transfer ramp system, wherein the extendable arm is operably coupled to the transfer ramp. 61. The method of any of examples 48-60 wherein positioning the transfer ramp includes rotating an extendable arm of the transfer ramp system after the transfer ramp makes contact with the platform or the surface of the other structure, wherein the extendable arm is operably coupled to the transfer ramp. 62. The method of any of examples 48-61 wherein positioning the transfer ramp includes conforming the transfer ramp to an arrangement of the platform and the surface of the other structure. 63. The method of any of examples 48-62 wherein positioning the transfer ramp includes, after the transfer ramp makes contact with the platform or the surface of the other structure, adjusting a height of the platform toward a height of the surface of the other structure. 64. The method of any of examples 48-63, further comprising transferring cargo across a top surface of the transfer ramp while the transfer ramp bridges the gap. 65. The method of any of examples 48-64, further comprising, after positioning the transfer ramp, raising the transfer ramp, and wherein transitioning the transfer ramp includes transitioning the transfer ramp from the deployed orientation to the stowed orientation. 66. The method of example 65 wherein transitioning the transfer ramp from the deployed orientation to the stowed orientation includes actuating a strut of the transfer ramp system operably coupled to the transfer ramp. 67. The method of example 66 wherein actuating the strut includes venting air from the strut such that the strut retracts. 68. The method of any of examples 65-67 wherein transitioning the transfer ramp from the deployed orientation to the stowed orientation includes extending an extendable arm operably coupled to the transfer ramp such that a force is exerted on the transfer ramp via a cable operably coupled to the transfer ramp. 69. The method of any of examples 65-68 wherein transitioning the transfer ramp from the deployed orientation to the stowed orientation includes manually rotating the transfer ramp. 70. A method of operating a transfer ramp system (a) mounted to a vehicle or trailer and (b) including a transfer ramp usable to transfer cargo between (i) the vehicle or trailer and (ii) another structure separate from the vehicle or trailer, the method comprising: deploying the transfer ramp, wherein deploying the transfer ramp includes moving the transfer ramp between— a stowed position beneath frame rails of the vehicle or trailer, and an extended position above a platform of the vehicle or trailer, and wherein moving the transfer ramp includes controlling, via a microcontroller of the transfer ramp system, one or more actuators of the transfer ramp system such that the transfer ramp is moved through a gap between the platform and the other structure. 71. The method of example 70 wherein deploying the transfer ramp includes monitoring feedback received from the one or more actuators and controlling the one or more actuators based at least in part on the feedback. 72. The method of example 71 wherein monitoring the feedback includes monitoring one or more potentiometer readings corresponding to the one or more actuators. 73. The method of any of examples 70-72 wherein moving the transfer ramp includes controlling, via the microcontroller, the one or more actuators such that the transfer ramp is moved between the stowed position and the extended position according to a predetermined path. 74. The method of example 73 wherein controlling the one or more actuators such that the transfer ramp is moved between the stowed position and the extended position according to the predetermined path includes controlling the one or more actuators based, at least in part, on feedback received from the one or more actuators. 75. The method of example 74 wherein the feedback includes potentiometer readings corresponding to the one or more actuators. 76. The method of any of examples 73-75 wherein the predetermined path extends through the gap. 77. The method of any of examples 73-76 wherein the predetermined path extends from the stowed position to the extended position. 78. The method of any of examples 70-77 wherein deploying the transfer ramp includes navigating the transfer ramp through a space beneath the platform and above a liftgate of the vehicle or trailer. 79. The method of any of examples 70-78 wherein deploying the transfer ramp includes extending the transfer ramp at least partially through a space beneath the platform, rotating the transfer ramp such that a pitch of the transfer ramp is increased, and extending the transfer ramp with the increased pitch through the gap. 80. The method of any of examples 70-79 wherein deploying the transfer ramp includes deploying the transfer ramp in response to user input. 81. The method of any of examples 70-80, further comprising rotating the transfer ramp between a stored arrangement and a deployed arrangement. 82. The method of example 81 wherein rotating the transfer ramp between the stored arrangement and the deployed arrangement includes rotating the transfer ramp such that a top surface of the transfer ramp is more aligned with a top surface of the platform than the top surface of the transfer ramp while the transfer ramp is in the stored arrangement. 83. The method of example 81 or example 82 wherein rotating the transfer ramp includes releasing the transfer ramp from a latch. 84. The method of any of examples 81-83 wherein rotating the transfer ramp includes: manually releasing the transfer ramp from a latch; pivoting the transfer ramp from the stored arrangement toward the deployed arrangement; and/or latching the transfer ramp in a second latch such that the transfer ramp is releasably retained in the deployed arrangement. 85. The method of any of examples 81-83 wherein rotating the transfer ramp includes: manually releasing the transfer ramp from a latch; pivoting the transfer ramp from the deployed arrangement toward the stored arrangement; and/or latching the transfer ramp in a second latch such that the transfer ramp is releasably retained in the stored arrangement. 86. The method of any of examples 70-85, further comprising moving the transfer ramp between (i) the extended position and (ii) a deployed position in which the transfer ramp spans the gap between the platform and the other structure. 87. The method of example 86 wherein moving the transfer ramp between the extended position and the deployed position includes moving the transfer ramp toward the deployed position in which (a) a first portion of the transfer ramp is positioned above and contacts a top surface of the platform and/or (b) a second portion of the transfer ramp is positioned above and contacts a top surface of the other structure. 88. The method of example 86 or example 87 wherein moving the transfer ramp between the extended position and the deployed position includes controlling, via the microcontroller, the one or more actuators such that the transfer ramp is moved according to a second predetermined path. 89. The method of example 88 wherein moving the transfer ramp according to the second predetermined path includes retracting the transfer ramp toward the platform such that the transfer ramp is moved from the extended position to the deployed position. 90. The method of example 88 wherein moving the transfer ramp according to the second predetermined path includes extending the transfer ramp away from the platform such that the transfer ramp is moved from the deployed position to the extended position. 91. The method of any of examples 86-90 wherein moving the transfer ramp between the extended position and the deployed position includes moving the transfer ramp in response to user input. 92. The method of any of examples 86-91 wherein moving the transfer ramp between the extended position and the deployed position includes moving the transfer ramp toward the deployed position, and wherein moving the transfer ramp toward the deployed position includes retracting the transfer ramp toward the platform and the other structure beyond initial contact of the transfer ramp with the platform or the other structure. 93. The method of example 92 wherein retracting the transfer ramp toward the platform and the other structure beyond the initial contact includes bending the transfer ramp to conform to an alignment between the platform and the other structure. 94. The method of example 92 or example 93 wherein retracting the transfer ramp toward the platform and the other structure beyond the initial contact includes adjusting a height of the platform and/or a height of a top surface of the other structure. 95. The method of any of examples 70-94, further comprising retracting the transfer ramp between the extended position and the stowed position, wherein retracting the transfer ramp including controlling, via the microcontroller, the one or more actuators such that the transfer ramp is moved (a) between the extended position and the stowed position according to a second predetermined path and (b) through the gap between the platform and the other structure. 96. The method of example 95 wherein retracting the transfer ramp according to the second predetermined path includes retracting the transfer ramp through the gap, rotating the transfer ramp such that a pitch of the transfer ramp is decreased, and retracting the transfer ramp with the decreased pitch through a space beneath the platform. 97. The method of example 95 or example 96 wherein retracting the transfer ramp includes retracting the transfer ramp in response to user input. 98. The method of any of examples 95-97 wherein retracting the transfer ramp includes seating the transfer ramp in one or more notches of a stowage stop positioned under the frame rails. 99. The method of any of examples 70-98 wherein the gap has a width of 52 cm or less. 100. The method of any of examples 70-99 wherein the gap has a width of 38 cm or less. 101. The method of any of examples 70-100 wherein the gap has a width of 26 cm or less.

D. Conclusion

(180) From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments of the technology. The above detailed descriptions of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology as those skilled in the relevant art will recognize. For example, although steps are presented in a given order above, alternative embodiments may perform steps in a different order. Furthermore, the various embodiments described herein may also be combined to provide further embodiments.

(181) To the extent any material incorporated herein by reference conflicts with the present disclosure, the present disclosure controls. Where the context permits, singular or plural terms may also include the plural or singular term, respectively. Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or

more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. Furthermore, as used herein, the phrase “and/or” as in “A and/or B” refers to A alone, B alone, and both A and B. Additionally, the terms “comprising,” “including,” “having,” and “with” are used throughout to mean including at least the recited feature(s) such that any greater number of the same features and/or additional types of other features are not precluded. Moreover, as used herein, the phrases “based on,” “depends on,” “as a result of,” and “in response to” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both condition A and condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on” or the phrase “based at least partially on.”
(182) Spatially relative terms, such as “beneath,” “below,” “over,” “under,” “above,” “upper,” “top,” “bottom,” “left,” “right,” “center,” “middle,” “forward,” “away,” and the like, are used herein for ease of description to describe one element or feature's relationship relative to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device or system in use or operation in addition to the orientation depicted in the figures. For example, if a device or system in the figures is rotated or turned over, elements described as “below” or “beneath” or “under” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary terms “below” and “under” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated ninety degrees or at other orientations) and the spatially relative descriptors used herein are interpreted accordingly. In addition, it will also be understood that when an element is referred to as being “between” two other elements, it can be the only element between the two other elements, or one or more intervening elements may also be present.
(183) From the foregoing, it will also be appreciated that various modifications may be made without deviating from the disclosure or the technology. For example, one of ordinary skill in the art will understand that various components of the technology can be further divided into subcomponents, or that various components and functions of the technology may be combined and integrated. In addition, certain aspects of the technology described in the context of particular embodiments may also be combined or eliminated in other embodiments. Furthermore, although advantages associated with certain embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. A transfer ramp system mountable to or under frame rails of a vehicle or trailer, the transfer ramp system comprising: a transfer ramp; one or more extendable arms operably coupled to the transfer ramp; one or more extension actuators operably coupled to the one or more extendable arms and configured to extend or retract the one or more extendable arms; and one or more rotation actuators operably coupled to the one or more extendable arms and configured to pivotably move the one or more extendable arms, wherein, when the transfer ramp system is mounted to or under the frame rails, the one or more extension actuators and the one or more rotation actuators are controllable to manipulate the one or more extendable arms to move the transfer ramp (a) from a position beneath a platform of the vehicle or trailer, (b) upwards and fully through a gap between the platform and another structure physically separate from the vehicle or trailer, and (c) to a position above the platform and a top surface of the other structure such that the transfer ramp is usable to transfer

cargo between the platform and the top surface.

2. The transfer ramp system of claim 1 wherein: the one or more extendable arms include a first extendable arm extending generally along a first longitudinal axis and a second extendable arm different from the first extendable arm, wherein the second extendable arm extends generally along a second longitudinal axis; the one or more extension actuators include a first actuator and a second actuator; the one or more rotation actuators include a third actuator and a fourth actuator; the first actuator is configured to extend or retract the first extendable arm generally along the first longitudinal axis; the third actuator is configured, at least while the first actuator is stationary, to pivot the first extendable arm and adjust a pitch of the first extendable arm; the second actuator is configured to extend or retract the second extendable arm generally along the second longitudinal axis; and the fourth actuator is configured, at least while the second actuator is stationary, to pivot the second extendable arm and adjust a pitch of the second extendable arm.

3. The transfer ramp system of claim 1, further comprising a microcontroller configured to control the one or more extension actuators and the one or more rotation actuators to manipulate the one or more extendable arms to move the transfer ramp (a) from the position beneath the platform, (b) through the gap, and (c) to the position above the platform, according to a predetermined path.

4. The transfer ramp system of claim 1 wherein the transfer ramp is configured to pivotably move between (a) a stowed orientation for stowage beneath the platform and (b) a deployed orientation in which a top surface of the transfer ramp is generally parallel to a top surface of the platform.

5. The transfer ramp system of claim 1, further comprising a ramp latching system configured to: releasably retain the transfer ramp in a first orientation for stowage beneath the platform; or releasably retain the transfer ramp in a second orientation in which a top surface of the transfer ramp is generally parallel with a top surface of the platform.

6. The transfer ramp system of claim 1 wherein: the transfer ramp has a length measured in a direction generally along a longitudinal axis of the vehicle or trailer; the length is shorter than a distance between (i) a platform of the vehicle or trailer and (ii) the ground upon which the vehicle or trailer rests; and the gap between the platform and the other structure, as measured in the direction generally along the longitudinal axis of the vehicle or trailer, is smaller than the length of the transfer ramp.

7. The transfer ramp system of claim 6 wherein the one or more extension actuators and the one or more rotation actuators are controllable to manipulate the one or more extendable arms to move an entirety of the length of the transfer ramp (i) upwards and through the gap between the platform and the other structure physically separate from the vehicle or trailer, and (ii) to the position above the platform and the top surface of the other structure.

8. The transfer ramp system of claim 1 wherein the one or more extension actuators and the one or more rotation actuators are controllable to manipulate the one or more extendable arms to move the transfer ramp to the position above the platform and the top surface of the other structure such that the transfer ramp, while in a deployed orientation that is generally parallel to the platform and the top surface of the other structure, rests on top of the platform, on top of the top surface, or a combination thereof.

9. The transfer ramp system of claim 1 wherein the one or more rotation actuators are further configured to pivotably move the one or more extension actuators.

10. A transfer ramp system mountable beneath a top platform at a rear of a vehicle or trailer, the transfer ramp system comprising: a transfer ramp; and at least one extendable arm attached to the transfer ramp and usable to extend or retract the transfer ramp, wherein, when the transfer ramp system is mounted beneath the top platform, the transfer ramp is moveable, using the at least one extendable arm, between (a) a stowed position in which the transfer ramp is positioned beneath the top platform and (b) a deployed position in which the transfer ramp is positioned above the top platform and another structure such that the transfer ramp is usable to bridge a gap between the top platform and another structure for moving cargo between (i) the vehicle or trailer and (ii) the other

structure, and wherein, in the deployed position, the transfer ramp is in a deployed orientation in which a top surface of the transfer ramp is generally parallel to a top surface of the top platform.

11. The transfer ramp system of claim 10 wherein the transfer ramp system is mountable beneath the top platform at the rear of the vehicle or trailer such that, when the transfer ramp is moved between the stowed position and the deployed position, the transfer ramp is navigated through a space below the top platform and above a liftgate of the vehicle or trailer.

12. The transfer ramp system of claim 10, further comprising at least one actuator configured to extend or retract the at least one extendable arm to extend or retract, respectively, the transfer ramp.

13. The transfer ramp system of claim 12 wherein the at least one actuator includes a first actuator usable to extend or retract the at least one extendable arm, and wherein the transfer ramp system further comprises a second actuator usable to rotate the at least one extendable arm.

14. The transfer ramp system of claim 12 wherein: the at least one extendable arm includes a first extendable arm extending generally along a first longitudinal axis and a second extendable arm different from the first extendable arm, wherein the second extendable arm extends generally along a second longitudinal axis; the at least one actuator includes a first actuator and a second actuator; the transfer ramp system further comprises a third actuator and a fourth actuator; the first actuator is usable to extend or retract the first extendable arm generally along the first longitudinal axis; the third actuator is usable to rotate the first extendable arm and adjust a pitch of the first extendable arm; the second actuator is usable to extend or retract the second extendable arm generally along the second longitudinal axis; and the fourth actuator is usable to rotate the second extendable arm and adjust a pitch of the second extendable arm.

15. The transfer ramp system of claim 10, further comprising: at least one actuator configured to manipulate the at least one extendable arm; and a microcontroller configured to control the at least one actuator to move the transfer ramp between the stowed position and the deployed position.

16. The transfer ramp system of claim 15 wherein the microcontroller is configured to control the at least one actuator to move the transfer ramp between the stowed position and the deployed position generally along a predetermined path.

17. The transfer ramp system of claim 16 wherein the microcontroller is configured to control the at least one actuator to move the transfer ramp along the predetermined path based, at least in part, on feedback received from the at least one actuator.

18. The transfer ramp system of claim 16 wherein the predetermined path is generally S-shaped such that, as the transfer ramp is moved between the stowed position and the deployed position, the transfer ramp is navigated under the top platform, through the gap between the top platform and the other structure, and above the top platform and the other structure.

19. The transfer ramp system of claim 10 wherein the transfer ramp is configured to rotate between (i) a stowed orientation for stowage beneath the top platform and (ii) the deployed orientation.

20. The transfer ramp system of claim 19, further comprising a ramp latching system configured to releasably hold the transfer ramp in the stowed orientation.

21. The transfer ramp system of claim 19, further comprising a ramp latching system, wherein the ramp latching system includes: a first latch configured to releasably hold the transfer ramp in the stowed orientation; and a second latch configured to releasably hold the transfer ramp in the deployed orientation.

22. A cargo transferring system mountable to or under frame rails of a vehicle or trailer, the cargo transferring system comprising: a bridge deck; one or more extendable arms operably coupled to the bridge deck; and one or more actuators configured to extend and retract the one or more extendable arms, wherein, during operation of the cargo transferring system when mounted to or under the frame rails of the vehicle or trailer, the one or more actuators are controllable to transition the bridge deck between a stowed position under the frame rails, a transition position in which the one or more actuators are controllable to maneuver the bridge deck from beneath the platform of the vehicle or trailer, upwards and through a space between the vehicle or trailer and the other

structure, and to a position above the platform of the vehicle or trailer, and a deployed position in which a top sruface of the bridge deck is generally parallel to a top surface of a platform of the vehicle or trailer and is usable to transfer cargo between (i) the platform of the vehicle or trailer and (ii) a surface of another structure different from the vehicle or trailer.
23. The cargo transferring system of claim 22 wherein the bridge deck includes a length measured in a direction generally along a longitudinal axis of the vehicle or trailer, and wherein the space between the vehicle or trailer and the other structure, measured in the direction generally along the longitudinal axis of the vehicle or trailer, is less than the length of the bridge deck.
24. The cargo transferring system of claim 22 wherein: the one or more actuators are one or more first actuators configured to extend and retract the one or more extendable arms generally along one or more longitudinal axes of the one or more extendable arms; and the cargo transferring system further comprises one or more second actuators usable to adjust a pitch of the one or more extendable arms.
25. The cargo transferring system of claim 22 wherein the bridge deck is configured to rotate relative to the one or more extendable arms such that a top surface of the bridge deck is alignable with a top surface of the platform.
26. The cargo transferring system of claim 22, further comprising a ramp latching system configured to: releasably retain the bridge deck in a first orientation for stowage beneath the frame rails; and/or releasably retain the bridge deck in a second orientation in which a top surface of the bridge deck is generally parallel with a top surface of the platform.
27. The cargo transferring system of claim 22, further comprising a microcontroller configured to control movement of the bridge deck between the stowed position and the deployed position.
28. The cargo transferring system of claim 27 wherein the microcontroller is configured to control the movement of the bridge deck such that the bridge deck is moved between the stowed position and the deployed position according to a predetermined path.
29. The cargo transferring system of claim 28 wherein the microcontroller is configured to, based at least in part on feedback received from the one or more actuators, control the movement of the bridge deck according to the predetermined path.
30. The cargo transferring system of claim 28 wherein the predetermined path is generally S-shaped and extends between a point beneath the frame rails and a point above the platform.
31. A method of operating a transfer ramp system (a) mounted to a vehicle or trailer and (b) including a transfer ramp usable to transfer cargo between (i) the vehicle or trailer and (ii) another structure separate from the vehicle or trailer, the method comprising: deploying the transfer ramp, wherein deploying the transfer ramp includes moving the transfer ramp between a stowed position beneath frame rails of the vehicle or trailer, and an extended position above a platform of the vehicle or trailer, and wherein moving the transfer ramp includes controlling, via a microcontroller of the transfer ramp system, one or more actuators of the transfer ramp system such that the transfer ramp is moved upwards and through a gap between the platform and the other structure, thereby positioning the transfer ramp at a location above the platform and the other structure.
32. The method of claim 31, further comprising moving the transfer ramp between (i) the extended position and (ii) a deployed position in which the transfer ramp spans the gap between the platform and the other structure.
33. The method of claim 31 wherein: deploying the transfer ramp includes monitoring feedback received from the one or more actuators; moving the transfer ramp includes controlling, via the microcontroller, the one or more actuators such that the transfer ramp is moved between the stowed position and the extended position according to a predetermined path; and controlling the one or more actuators includes controlling the one or more actuators based, at least in part, on the feedback.
34. The method of claim 31, further comprising rotating the transfer ramp between a stored arrangement and a deployed arrangement in which a top surface of the transfer ramp is more

aligned with a top surface of the platform than in the stored arrangement.

35. The method of claim 31 wherein: the transfer ramp has a length measured in a direction generally along a longitudinal axis of the vehicle or trailer; the length is shorter than a distance between (i) a platform of the vehicle or trailer and (ii) the ground upon which the vehicle or trailer rests; and the gap between the platform and the other structure, as measured in the direction generally along the longitudinal axis of the vehicle or trailer, is smaller than the length of the transfer ramp.

36. The transfer ramp system of claim 10 wherein: the transfer ramp has a length measured in a direction generally along a longitudinal axis of the vehicle or trailer; and the length is shorter than a distance between (i) a platform of the vehicle or trailer and (ii) the ground upon which the vehicle or trailer rests.